Data, information and units of storage

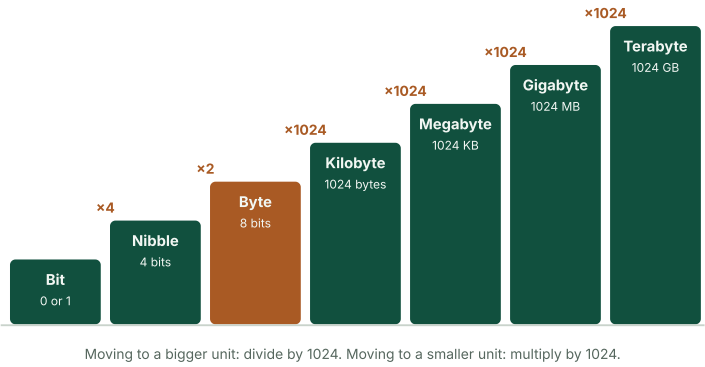
L01 FACT FILE

Specification: students should be able to

- Describe the difference between information and data
- Describe how data is stored in bit, nibble, byte, kilobyte, megabyte, gigabyte and terabyte
- Explain how characters are represented in binary using a character set such as ASCII
- Calculate conversions between units of storage

CCEA fact file: Digital Data 01: Representing Data

Key words: data, information, context, information system, binary, bit, nibble, byte, character set, ASCII, kilobyte, megabyte, gigabyte, terabyte, storage capacity



Moving to a bigger unit: divide by 1024. Moving to a smaller unit: multiply by 1024.

Units of storage as a staircase from bit to terabyte, with the multiplier between each step

Data and information

Data is raw facts and figures that have no meaning on their own. The numbers 07, 21 and 14 are data: you cannot tell what they are about. Information is data that has been processed and given a context, so it now has meaning. 'Enniskillen Rugby Club U16 results: won by 7 points, lost by 21 points, won by 14 points' is information, because we now know what the numbers describe. A useful way to remember this is that information is data with meaning. The same data can become different information depending on its context. For example, 140526 might be a date of birth (14/05/26), a product number in a shop or a winning lottery ticket. An information system, which is hardware and software working together, accepts data as input, processes it and outputs information. A supermarket till scanning barcodes is collecting data; the weekly sales report printed for the manager is information.

Bits, nibbles and bytes

Computers store everything in digital form using binary, a number system that has only two digits, 0 and 1. This suits computers because their circuits are built from switches that are either off (0) or on (1). One binary digit is called a bit, and it is the smallest unit of storage. Four bits make a nibble, which is half a byte and can store one hexadecimal digit. Eight bits make a byte. One byte is enough to store a single character, such as a letter, a digit or a punctuation mark. So the town name 'Strabane' has eight characters, takes eight bytes and needs 64 bits. Bits on their own can only represent two states, but grouping them allows many more values: a byte has 256 different patterns, from 00000000 to 11111111.

Character sets and ASCII

When you press a key, the computer needs a way to turn that character into binary. It does this using a character set, which is an agreed list that gives every character its own unique binary code. ASCII (American Standard Code for Information Interchange) is a widely used character set. In ASCII the capital letter A is 01000001, B is 01000010 and C is 01000011, so the codes for letters go up in order. Lower case letters have different codes: a is 01100001. Because both the sending and receiving computers use the same character set, text typed on one computer appears correctly on another. This is why a computer uses character codes: it can only store and process binary, and a shared code means text can be stored, sent and displayed reliably. Modern systems often use larger character sets, such as Unicode, to include symbols and other alphabets.

Units of storage and conversions

Storage capacity, such as the size of a USB flash drive, a hard drive or a computer's memory, is measured in bytes and multiples of bytes. Because computers count in binary, each unit is 1024 (which is 2 to the power of 10) times the one before. There are 1024 bytes in a kilobyte, 1024 kilobytes in a megabyte, 1024 megabytes in a gigabyte and 1024 gigabytes in a terabyte. To convert to a bigger unit, divide by 1024; to convert to a smaller unit, multiply by 1024. Remember that bits and bytes are linked by 8, not 1024. For example, 2 GB is 2 × 1024 = 2048 MB, and 12 288 KB is 12 288 ÷ 1024 = 12 MB. In 'how many will fit' questions, change both values into the same unit first, then divide the storage capacity by the size of one file. Check your answer makes sense before moving on.

T Key terms from the mark schemes

Data
Raw facts and figures that have no meaning or context on their own.
Credited: unprocessed facts and figures

Information
Data that has been processed and given context so that it has meaning.
Credited: data with meaning

Bit
A binary digit: the smallest unit of digital data, either a 0 or a 1 (off or on).
Credited: binary digit

Nibble
A group of 4 bits (half a byte). One hexadecimal digit can be stored in a nibble.

Byte
A group of 8 bits. One keyboard character (for example a letter, a digit or a punctuation mark) takes up 1 byte.
Credited: 8 binary digits

Kilobyte (KB)
1024 bytes – enough for a short plain-text file.

Megabyte (MB)
1024 kilobytes – around 1 minute of MP3 audio.

Gigabyte (GB)
1024 megabytes – around 20 minutes of HD video.

Terabyte (TB)
1024 gigabytes – the capacity of a typical laptop or external hard drive.

ASCII
American Standard Code for Information Interchange: a character set that gives each letter, digit and punctuation mark a unique binary code.
Credited: character set

Information system
Hardware and software working together to take data as input, process it and produce information as output.

Exam focus

This lesson appears in almost every Unit 1 paper. Data and information with examples was worth 4 marks in 2019 Q2(a): one mark for each definition and one for each suitable example, and the information example had to show context. 2023 Q1 asked about bits, bytes and data in context. Unit questions are often short: how many bits are in a nibble and how many kilobytes are in a megabyte (2019 Q2(b)), matching bit, nibble and byte to their definitions (2021 Q4), and converting 2 GB into megabytes (2025 Q4(c), answer 2048 MB). Common mistakes include writing 1000 instead of 1024, mixing up bits and bytes, dividing when you should multiply, and giving an 'information' example that is still just numbers. Use the exact terms bit, byte and binary digit, and always show your working in calculations, because method marks can be awarded even if the final answer is wrong.

Unit 1 2025 Q4	[7]	Boot process, OS roles, storage conversion
Unit 1 2023 Q1	[5]	Bit and byte, data types, data in context
Unit 1 2021 Q4	[6]	Bits, nibble, byte; data types matching
Unit 1 2019 Q2(b)	[2]	Bits in a nibble; KB in an MB
Unit 1 2019 Q2(a)	[4]	Data and information with examples

? Check yourself

1. Using an example of each, describe the difference between data and information. [4]
2. Put these units in order from smallest to largest: megabyte, bit, terabyte, byte, nibble, kilobyte, gigabyte. [2]
3. (a) How many bits are in a nibble? (b) How many kilobytes are in a megabyte? (c) How many bytes are needed to store the word 'Fermanagh'? [3]

Answers: 1. Data is raw facts and figures with no meaning / context (1); suitable example, e.g. 18, 22, 19 (1); information is data that has been processed / given context so it has meaning (1); suitable example showing the context, e.g. 'Belfast maximum temperature on Monday was 18°C' (1). 2. bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte. 2 marks for all in correct order; 1 mark if only one item is out of place. 3. (a) 4 (1) (b) 1024 (1) (c) 9 bytes – 9 characters, 1 byte each (1).

Data types

L02
FACT FILE

Specification: students should be able to

- Identify the following data types: numeric (integer and real), date/time, character and string
- Explain why data must be defined as a particular data type before it is processed
- Select and justify the most appropriate data type for a given item of data

CCEA fact file: Digital Data 01: Representing Data

Key words: data type, numeric, integer, real, double, date/time, date format, character, string, character set, type check, currency, Boolean (Yes/No), leading zero

Field	Example value	Data type
Surname	Maguire	String
Age	15	Integer
Height (m)	1.62	Real
Form initial	B	Character
Date of birth	12/09/2010	Date/time
Phone	07700 900123	String

Phone numbers are strings: the leading 0 must be kept and no maths is done on them.

A pupil record with each field matched to its data type

What is a data type?

A data type tells the computer what kind of data an item is, for example a whole number, a date or some text. Data must be defined as a particular type before it is processed. This matters for three main reasons. First, it decides what processing is possible: numbers can be added and averaged, dates can be put in order or used to work out how many days are left, and text can be searched or joined together. Second, it helps the computer store data efficiently, because each type is stored in a particular way. Third, it supports validation. If a field is set to integer, a type check will reject an entry such as 'twelve', which helps stop errors getting into the system. Choosing the right type when a database, spreadsheet or program is set up saves problems later.

Numeric data: integer and real

Numeric data is data that you can do calculations with. The specification splits it into two types. An integer is a whole number with no decimal or fractional part. It can be positive or negative, so 28, 0 and -4 are all integers. Integers suit counts, such as the number of pupils in a class, tickets sold for a school show or a score of 51 in a test. A real number can have a decimal or fractional part, such as 1.62, 3.99 or -4.6. Real numbers can also be positive or negative, and the set of real numbers includes whole numbers too. Real is used for measurements and money, such as heights, temperatures and prices. In C#, the course programming language, int stores integers and double stores real numbers. In a database, money is often stored using a currency type, which is shown with a currency symbol and two decimal places.

Date/time, character and string

The date/time type stores dates, times or both. Because the computer knows the value is a date, it can sort records by date and calculate with them, such as finding everyone whose membership ends this month. Date/time also controls how the value is displayed. Software offers ready-made formats, for example a long date (Saturday, 12 September 2026), a medium date (12-Sep-2026) or a short date (12/09/2026), and similar long, medium and short time formats; users can also create their own. A character is one single letter, digit or symbol, such as a grade 'B' or 'Y' for yes. It is stored using its code from the character set, such as ASCII. A string is a sequence of characters, such as a name, address or postcode. Strings can vary in length. Any digits in a string are treated as text, so they cannot be used in calculations.

Choosing and justifying a data type

Many exam questions give you an item of data and ask for the most suitable type and a reason. Work through a few questions in order. Will calculations be done with it? If so it is numeric: real if it needs a decimal part, integer if not. Is it a date or time? Then use date/time. Is it always exactly one character? Then character. Everything else is a string. Watch for 'numbers' that are really text. A mobile phone number such as 07700 900123 must be a string, because storing it as an integer would remove the leading zero, the space would not be allowed and nobody ever adds phone numbers together. Postcodes such as BT92 1AB mix letters and digits, so they are strings too. A pupil's age is an integer, but their date of birth is date/time, because the age can be worked out from it and it stays correct as time passes.

Key terms from the mark schemes

Data type
The kind of data an item is (such as integer, real, date/time, character or string), which decides how it is stored and processed.

Real
Data type for numbers that can have a decimal or fractional part, positive or negative. Real numbers also include whole numbers.
`Credited: double / float / decimal`

Character
Data type for a single letter, digit or symbol, stored using its code from the character set.
`Credited: char`

Currency
A database data type for money values, shown with a currency symbol and two decimal places.

Type check
A validation check that makes sure the data entered is of the correct data type.

Integer
Data type for whole numbers, positive or negative, with no decimal or fractional part.
`Credited: int / whole number`

Date/time
Data type that stores dates and/or times so that they can be sorted, calculated with and shown in different formats.

String / text
Data type for a sequence of characters (letters, digits and symbols) of variable length.
`Credited: text`

Boolean / Yes-No
Field type that can only hold one of two possible values, for example Yes or No (True or False).
`Credited: logical`

Exam focus

Data types come up in most Unit 1 papers, often worth one mark per item. You may be asked to match sample values to data types (2021 Q4(c), 2025 Q5(a)), choose types for database fields (2019 Q3(b), 2024 Q3(a)) or justify a choice (2019 Q3(c)). In 2023 Q1 the data types question was part of a short-answer question on data. A 'state' question needs just the type name. A 'describe' question needs what the type stores plus an example, for example 'Real stores numbers with a decimal part, such as a price of 2.85'. A justification must link to the data itself. Saying 'because it is a number' will not earn the mark; 'because it is a whole number used to count pupils' will. In database questions, Access names such as Currency, Short Text or Yes/No are also credited. The most frequent lost marks come from calling a phone number an integer.

`Unit 1 2025 Q5`	`[11]`	Data types, validation checks, data processing
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q3`	`[11]`	Field names and data types, database terms, key field, query design
`Unit 1 2023 Q1`	`[5]`	Bit and byte, data types, data in context
`Unit 1 2021 Q4`	`[6]`	Bits, nibble, byte; data types matching

Check yourself

1. State the most suitable data type for: (a) the number of pupils in a class (b) the price of a school lunch (c) a pupil's date of birth (d) a pupil's surname (e) a grade, e.g. A. [5]
2. Explain why a mobile phone number should be stored as a string and not an integer. [2]
3. Describe the difference between the integer and real data types. [2]

Answers: 1. (a) integer (1) (b) real / currency (1) (c) date/time (1) (d) string / text (1) (e) character (1). 2. A phone number starts with 0 and the leading zero would be lost if stored as an integer (1); it may contain spaces / a + sign, which integers cannot hold (1); it is never used in calculations (1). Max 2. 3. An integer is a whole number with no decimal / fractional part, e.g. 15 (1); a real number can have a decimal / fractional part, e.g. 15.75 (1).

Representing images: pixels, resolution, bitmap and vector

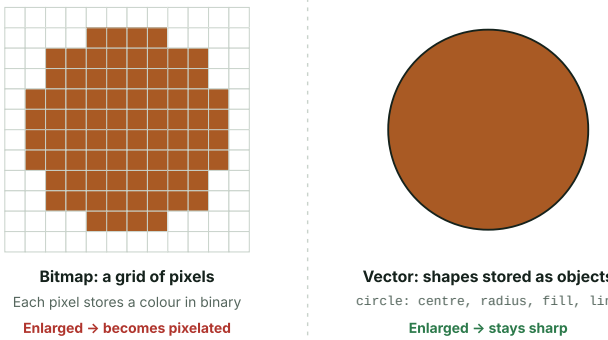
L03 FACT FILE

Specification: students should be able to

- Demonstrate understanding of how pixels are used in image representation
- Demonstrate understanding of how image resolution affects file size
- Describe how vector-based graphics and bitmap graphics are stored
- Describe the difference between vector-based and bitmap graphics
- Calculate the file size of a bitmap image from its resolution and colour depth

CCEA fact file: Digital Data 02: Representing Images

Key words: pixel, bitmap, vector, resolution, colour depth, file size, pixelation, metadata, coordinates, ppi / dpi, object, properties, scaling, SVG, 2-bit colour



Bitmap image as a grid of pixels compared with a vector circle stored as an object

Pixels and bitmap images

A pixel, short for picture element, is the smallest part of a digital image that can be shown and edited on a screen. A bitmap image is a grid of pixels, and each pixel can be a different colour. When all the pixels are seen together they make the complete picture. Each pixel's colour is stored as a binary code, and its place in the grid is given by coordinates, like (x, y) on a graph. When you take a photo on your phone or scan a page, you create a bitmap. To save a bitmap and display it again, the computer stores the colour code of every pixel plus metadata, which is data about the image such as its width, height and colour depth. Without the metadata, the computer would have a long list of colour codes but would not know where each row of the picture ends.

Colour depth

Colour depth is how many bits are given to each pixel to record its colour. The more bits per pixel, the more different colours the image can contain. One bit gives two possible codes, 0 and 1, so only two colours such as black and white. Two bits give four codes (00, 01, 10 and 11), so four colours. Every extra bit doubles the number of colours, so 8 bits give 256 colours and 24 bits give about 16.7 million colours. Most photographs use 24-bit colour, with 8 bits each for the amounts of red, green and blue. To work out how many bits you need, find the smallest power of 2 that is at least the number of colours. For example, an image with 5 colours needs 3 bits, because 2 bits only give 4 codes. More colours make an image more realistic but increase the file size.

Resolution and file size

Resolution describes how many pixels an image contains. It is usually written as two numbers, such as 1280 × 720: the first is the number of pixels across and the second is the number of pixels down. It can also be measured as pixels per inch (ppi) on a screen, or dots per inch (dpi) when printing. The higher the resolution, the more pixels there are, so the image is sharper and shows more detail. However, every pixel must be stored, so higher resolution means a larger file that takes more storage space and longer to load or upload. To estimate the size of the image data, multiply width by height by colour depth to get bits, then divide by 8 for bytes and by 1024 for kilobytes. A 10 × 10 image with 4 colours uses 10 × 10 × 2 = 200 bits, which is 25 bytes.

Vector graphics

Vector graphics are not made from a grid of pixels. Instead, the file stores a list of the objects that make up the picture, such as lines, curves, rectangles and circles, together with their properties. Typical properties are the position of the object, its size, its line colour, line thickness and fill colour. Because these objects are described mathematically, the computer recalculates and redraws them whenever the image is resized. This means a vector graphic can be enlarged as much as you like without losing quality or becoming pixelated. Each object can be selected and edited on its own, for example moving a shape or changing its colour. Vector files are usually smaller than bitmaps because they store a description of each shape rather than every pixel. They are ideal for logos, diagrams, maps and fonts, but not for photographs, which have too much colour detail. SVG is a common vector format.

Comparing bitmap and vector

The key difference is how each type is stored. A bitmap stores the colour of every pixel in a grid, while a vector stores objects and their properties. This leads to other differences. When a bitmap is enlarged, its pixels get bigger and the image becomes blocky, which is called pixelation; a vector is recalculated, so it stays sharp at any size. Bitmap file sizes grow with resolution and colour depth and can be very large, while vector files are usually smaller. A bitmap is edited pixel by pixel, whereas a vector is edited one whole object at a time. Bitmaps are best for photographs and scanned images where colours change gradually, and vectors are best for images built from shapes, such as a school crest that must look sharp both on a small badge and on a large sign at the school gate.

T Key terms from the mark schemes

Pixel
The smallest element of a digital picture that a screen can show or an editor can change. Every pixel holds one colour, recorded as a binary code.
`Credited: picture element`

Vector graphic
An image stored as a list of objects (lines, curves and shapes) with properties such as position, size, line colour and fill colour. Can be enlarged without loss of quality. Example file type: SVG.
`Credited: vector-based graphic`

Colour depth
The number of bits used to store the colour of each pixel. n bits give 2^n colours.
`Credited: bits per pixel`

Bitmap graphic
An image made of a grid of pixels, each of which can be a different colour. The colour code of every pixel is stored, plus metadata such as the dimensions. Pixelation occurs when it is enlarged. Example file type: JPEG.
`Credited: raster image`

Resolution
The number of pixels in an image (width × height) or per inch. More pixels give a sharper image but a bigger file size.
`Credited: image resolution`

Pixelation
The blocky appearance of a bitmap image when it is enlarged too much, because the individual pixels become visible.

PPI / DPI
Pixels per inch (screen) / dots per inch (printing): measures of resolution.

Metadata
Data about the data, e.g. the width, height and colour depth of an image, stored so that the image can be rebuilt for display.

Exam focus

Bitmap and vector graphics appear regularly. 2023 Q2(a) and 2025 Q2(a) asked pupils to describe differences or decide which statements apply to bitmap or vector graphics. For a 'describe two differences' question, each mark normally needs a comparison, such as 'a bitmap is stored as a grid of pixels whereas a vector is stored as objects and their properties'. Resolution and file size were tested in 2019 Q4(a) and 2023 Q2(b): you must say that more pixels give a sharper image but a larger file. 2019 Q4(a)(iii) asked for units of resolution, so learn ppi and dpi. Defining a pixel needs 'smallest unit of a digital image'; simply saying 'a dot' is too vague. In file size calculations, write the formula, show each step and give units. Common mistakes are forgetting to divide by 8 for bytes and using the number of colours instead of the number of bits.

Unit 1 2025 Q2	[6]	Bitmap vs vector; sound file size calculation
Unit 1 2023 Q2	[10]	Bitmap vs vector, resolution, buffering, sample rate, ADC, compression
Unit 1 2019 Q4(a)	[6]	Image file type, compression, resolution, ppi/dpi

? Check yourself

1. Define the following terms: (a) pixel (b) image resolution. [2]
2. Explain how increasing the resolution of an image affects its quality and its file size. [3]
3. Describe how a vector graphic is stored. [2]

Answers: 1. (a) The smallest unit / element of a digital image that can be shown on screen and edited (1). (b) The number of pixels in an image, given as width × height / pixels per inch (1). 2. More pixels are used (1); so the image is sharper / shows more detail / higher quality (1); but more data must be stored so the file size increases (1). 3. Stored as a list of objects / shapes such as lines, curves and circles (1); with their properties, e.g. position / coordinates, size, line colour, fill colour, line thickness (1).

Representing sound; buffering and streaming

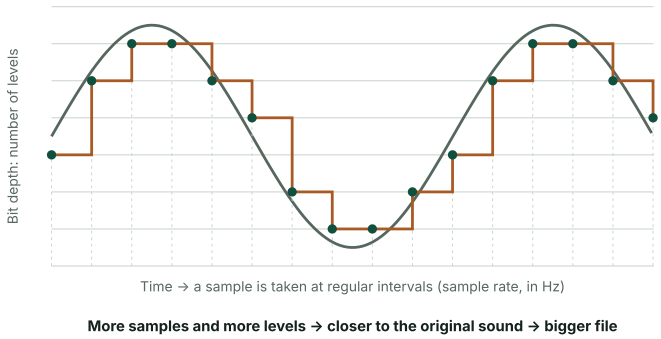
L04 FACT FILE

Specification: students should be able to

- Explain the need for analogue-to-digital conversion in sound recording
- Describe factors that affect sound quality when recording sound, including sample rate, bit depth and bit rate
- Calculate the bit rate and file size of a sound recording
- Demonstrate understanding of how buffering and streaming are used to support the transfer of moving image files

CCEA fact file: Digital Data 03: Representing Sound

Key words: analogue signal, digital, microphone, analogue-to-digital converter (ADC), sampling, sample, amplitude, sample rate, hertz (Hz), bit depth, bit rate, channels (mono/stereo), file size, streaming, downloading, buffer, buffering



A sound wave sampled at regular intervals, with sample rate on the time axis and bit depth as the number of levels

How a computer records sound

Sound is a wave of vibrations. It is an analogue signal, which means it changes smoothly and continuously. A computer can only work with digital data, which is made of binary 0s and 1s, so the sound has to be converted before it can be stored. First a microphone picks up the sound and turns it into an analogue electrical signal. Then an analogue-to-digital converter (ADC) measures the height of the signal at regular, equal time intervals. Each measurement is called a sample. The ADC rounds each sample to the nearest whole-number level and stores that number in binary. The finished recording is just a long list of binary numbers. Because the wave is only measured at set moments, anything the wave does between two samples is not recorded, so a digital recording is always an approximation of the original sound. How good that approximation is depends on the sample rate and the bit depth.

Sample rate, bit depth and bit rate

Sample rate is the number of samples taken each second. It is measured in hertz (Hz), so 44 100 Hz means 44 100 samples per second, which is the standard for CDs. More samples per second follow the shape of the wave more closely, so the sound is more like the original. Bit depth is how many bits are given to each individual sample. With 8 bits there are 256 possible levels, but with 16 bits there are 65 536, so each sample is rounded much less and is stored more accurately. Bit rate is the number of bits processed for every second of sound. For one channel it is the sample rate multiplied by the bit depth, and for stereo you multiply by two because there is a left and a right channel. Increasing any of these three improves the quality, but it also increases the file size, so the recording takes up more storage and takes longer to send.

Working out file sizes

To find the size of an uncompressed sound file, multiply the sample rate by the bit depth by the number of channels by the length of the recording in seconds. The answer is in bits. Divide by 8 to change bits into bytes, then by 1024 to change bytes into kilobytes, and by 1024 again to get megabytes. For example, a 30-second mono clip recorded at 11 025 Hz with an 8-bit depth needs 11 025 × 8 × 30 = 2 646 000 bits. Dividing by 8 gives 330 750 bytes, which is about 323 KB. Always write the formula down first, then put in the numbers, and keep track of the units at each step. A very common mistake is forgetting to change minutes into seconds, so a 2-minute track must use 120 seconds. Another is forgetting to divide by 8. Compressed formats such as MP3 store much less than this, because some data is removed.

Streaming and buffering

Video files are very large, so downloading a whole film before watching it can take a long time. Streaming solves this. The video is sent over the internet in small pieces and starts playing as soon as the first pieces arrive. The user watches in real time while the rest is still being received, and the file is not normally saved permanently on the device. To make streaming smooth, the device uses a buffer. This is an area of memory that stores the next part of the video before it is needed. While one section is playing, the following section is downloaded into the buffer. If the internet connection slows down or drops briefly, the video keeps playing from the buffer, so the viewer does not notice. Only if the buffer empties completely does the video stop and show a loading icon. Streaming needs a constant connection, which can be a problem in rural areas with weak broadband or mobile signal.

T Key terms from the mark schemes

Analogue signal
A signal that varies smoothly and continuously, representing a physical quantity such as a sound wave.
`Credited: continuously varying signal`

Analogue-to-digital converter (ADC)
A device that converts an analogue signal, such as sound from a microphone, into digital (binary) data by sampling it at regular intervals.
`Credited: U1 2022 Q1, 2023 Q2`

Sampling
Measuring the level (amplitude) of an analogue signal at regular time intervals; each reading is rounded to a whole-number level and stored in binary.

Sample rate
The number of samples of a sound taken each second, measured in hertz (Hz). CD audio uses 44 100 Hz (44.1 kHz).
`Credited: U1 2023 Q2, 2025 Q2; sampling frequency`

Bit depth
How many bits are allocated to every sound sample. More bits give more possible levels, so each sample is recorded more accurately.
`Credited: U1 2024 Q7, 2025 Q2`

Bit rate
The number of bits of data processed for each second of sound, usually in kilobits per second; for one channel it equals sample rate × bit depth.
`Credited: U1 2019 Q1, 2022 Q1`

Streaming
Playing video or audio over the internet while it is still being received, without waiting to download and permanently store the whole file.
`Credited: U1 2019 Q4`

Buffer / buffering
An area of memory that temporarily stores the next part of a video or song before it is played, so playback continues if the connection slows or drops briefly.
`Credited: U1 2019 Q4, 2021 Q11, 2022 Q1, 2023 Q2`

Stereo
Sound recorded on two channels (left and right); it needs twice as much data as the same sound in mono (one channel).

Exam focus

This lesson comes up almost every year. Bit rate has appeared as a multiple-choice question (2019 Q1(j), 2022 Q1). The purpose of an ADC was tested in 2022 Q1 and 2023 Q2(e): say it converts the analogue signal into digital (binary) data by sampling it at regular intervals. Sample rate was asked in 2023 Q2(d) and bit depth in 2024 Q7(b); for each, give a clear definition and then say how increasing it affects both quality and file size. In 2025 Q2(b)–(c) pupils had to calculate the size of a sound file, so show every step: the formula, the multiplication, then the conversion from bits to bytes and KB. Streaming and buffering were tested in 2019 Q4(b), 2022 Q1 and 2023 Q2(c). Marks come from saying that the buffer is temporary memory, that it stores the next part of the video in advance, and that playback continues from it if the connection slows. Do not mix up bit depth (bits per sample) with bit rate (bits per second).

Unit 1 2025 Q2	[6]	Bitmap vs vector; sound file size calculation
Unit 1 2024 Q7	[5]	Compression, video formats, bit depth
Unit 1 2023 Q2	[10]	Bitmap vs vector, resolution, buffering, sample rate, ADC, compression
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup
Unit 1 2021 Q11	[8]	Key term matching

? Check yourself

1. Explain why analogue-to-digital conversion is needed when recording sound on a computer. [2]
2. State the purpose of an analogue-to-digital converter (ADC). [1]
3. What is meant by sample rate? State the unit it is measured in. [2]

Answers: 1. Sound is an analogue signal / continuous wave (1) / computers can only store and process digital (binary) data, so the sound must be converted (1). 2. To convert an analogue signal (e.g. from a microphone) into digital / binary data (by sampling it) (1). 3. The number of samples taken per second (when recording sound) (1) / measured in hertz (Hz) (1).

Data portability, file formats and compression

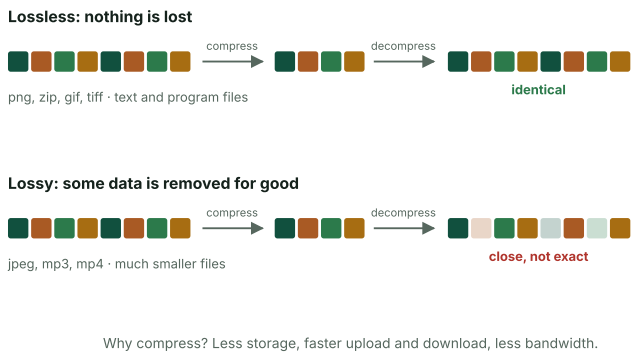
L05
FACT FILE

Specification: students should be able to

- Demonstrate understanding of data portability and the following file formats that support it: jpeg, tiff, png, pict, gif, txt, csv, rtf, mp3, mp4, midi, mpeg, avi, pdf, wav and wma
- Choose and justify a suitable file format for a given purpose
- Demonstrate understanding of the need for data compression
- Describe lossy and lossless compression and give examples of each

CCEA fact file: Digital Data 04: Portability

Key words: data portability, interoperability, file format, file extension, export, import, compression, lossy compression, lossless compression, zip, jpeg, png, gif, tiff, csv, rtf, pdf, midi



Lossless compression restores every block of data; lossy compression permanently removes some

Data portability and file formats

Data portability means being able to move data from one computer, device or application to another without having to type it in again. Mark schemes also accept the words interoperability or compatibility. Portability depends on file formats. A file format is the agreed way that data is laid out inside a file, and the file extension, such as .jpg or .csv, shows which format is used. If data is saved in a standard format, many different programs on many different systems can open it. For example, a youth football club in Lisburn might keep its members in a database, export them as a csv file and send that file to the league, which opens it in a spreadsheet. Nothing is retyped, so time is saved and no new errors are introduced. Choosing the right format matters. If you pick a format that the other person cannot open, or one that loses quality or formatting, the data is not truly portable.

Image, text and document formats

Jpeg is the usual format for photographs. It uses lossy compression, and the user can choose how much, so files are small and load quickly on web pages. Png uses lossless compression, keeps every detail and supports transparent backgrounds, which makes it good for logos. Gif can only use 256 colours, but it is lossless and can store a series of images that play one after another as a simple animation. Tiff keeps very high quality and moves easily between Windows and Apple computers, so it is used for printing and scanning, but files are large. Pict is Apple's own graphics format and can hold bitmap and vector images. For text, txt holds plain text with no formatting at all. Csv stores table data with commas between the fields. Rtf keeps basic formatting such as fonts and bold and can be opened by most word processors. Pdf, created by Adobe, keeps fonts, images and layout exactly the same on every device, and is usually read-only.

Sound and video formats

Wav stores sound without any compression, so the quality is high but the files are large, which makes it suitable for recording and editing. Mp3 compresses sound using lossy compression. It removes sounds that most people would not hear, giving a much smaller file that still sounds close to the original, so it is used for music and podcasts. Wma is Microsoft's compressed audio format, made for Windows Media Player. Midi is different from all the others. It does not store recorded sound at all. Instead it stores instructions such as which notes to play, on which instrument and for how long, so the files are tiny and can be passed between electronic keyboards and computers. For video, mp4 compresses both the pictures and the sound and can also hold subtitles and still images; it is widely used for streaming. Mpeg is named after the group of experts who created standards for compressing video. Avi is a Microsoft format that keeps audio and video in sync, but files are often large.

Data compression

Compression reduces the size of a file. Smaller files take up less storage space and can be uploaded, downloaded and emailed more quickly, which is important for web pages, streaming and attachments with size limits. There are two types. Lossy compression permanently removes some of the data, such as fine detail or colours that people are unlikely to notice. Files become much smaller, but some quality is lost and the original can never be fully recovered. Jpeg, mp3 and mp4 use lossy compression, and it suits photos, music and video. Lossless compression makes the file smaller without throwing any data away, so when the file is decompressed it is exactly the same as the original. Zipping a folder with a program such as WinZip is lossless, and so is png. Lossless compression does not shrink files as much, but it must be used for text documents, spreadsheets and programs, because losing even a small part would change the data or stop it working.

T Key terms from the mark schemes

Data portability
Being able to move data between computers, devices or software applications without typing it in again, usually by saving it in a standard file format.
Credited: interoperability; compatibility (U1 2021 Q9, 2022 Q11, 2025 Q3)

Compression
Reducing the size of a file so it needs less storage and can be uploaded or downloaded more quickly.
Credited: U1 2022 Q1, 2024 Q7

Lossless compression
Compression that reduces file size without losing any data, so the file can be restored exactly as it was, e.g. zip, png.
Credited: U1 2019 Q4, 2021 Q10, 2023 Q2

PDF
Portable Document Format, developed by Adobe: keeps a document's fonts, images and layout the same on any device; usually read-only.
Credited: U1 2025 Q3

TIFF
Tagged Image File Format: stores high-quality bitmap images that can move between Windows and Apple computers; files are larger than jpegs.
Credited: U1 2022 Q11

MIDI
Musical Instrument Digital Interface: stores and exchanges musical instructions (notes, instruments, timing) between electronic instruments and computers, not recorded sound.
Credited: U1 2021 Q10

File format
The standard way data is organised inside a file, shown by its file extension (e.g. .jpg, .csv), which determines which programs can open it.

Lossy compression
Compression that reduces file size by permanently removing some data, so some quality is lost and the original cannot be fully restored, e.g. jpeg, mp3.
Credited: U1 2019 Q4, 2021 Q10, 2023 Q2

CSV
Comma-separated values: a text format in which each line is a record and the fields are separated by commas; used to move table data between spreadsheets and databases.
Credited: U1 2021 Q2, 2022 Q10

JPEG
Image format using lossy compression whose level the user can control; small files, widely used for photographs on web pages.
Credited: U1 2021 Q10, 2022 Q11, 2025 Q3

WAV
A standard uncompressed audio format; high quality but large files.
Credited: U1 2025 Q3

RTF
Rich Text Format: a text document format with basic formatting that can be opened by different word processors and operating systems.

Exam focus

File formats appear almost every year. Multiple-choice questions on image and video formats came up in 2019 Q1(a)–(c), and 2021 Q10(a) and 2025 Q3(b) asked for formats and their uses. 2022 Q11 asked about portability and compared jpeg with tiff: credit came from saying jpeg uses lossy compression and gives small files for the web, while tiff keeps high quality for printing but makes large files. 2021 Q2(e) tested exporting database data to csv. The Chief Examiner reported that portability (2025 Q3(a)) was poorly answered, so learn a precise definition: moving data between computers or applications without re-entering it. For compression (2021 Q10(a)(ii), 2023 Q2(f), 2024 Q7(a)), give a reason such as faster upload or less storage, and when comparing lossy and lossless always state that lossy loses data permanently while lossless can be restored exactly. When asked to choose a format, name it and justify it with a feature that fits the task.

Unit 1 2025 Q3	[5]	Data portability and file formats
Unit 1 2024 Q7	[5]	Compression, video formats, bit depth
Unit 1 2023 Q2	[10]	Bitmap vs vector, resolution, buffering, sample rate, ADC, compression
Unit 1 2022 Q11	[4]	Data portability; JPEG and TIFF
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup

? Check yourself

1. What is meant by data portability? [2]
2. Name a suitable file format for each: (a) a photo for a website (b) a document that must look the same on every device (c) data exported from a database to a spreadsheet (d) an animated image with few colours. [4]
3. Describe the difference between a midi file and an mp3 file. [2]

Answers: 1. Being able to move data between computers / devices / software applications (1) / without it needing to be typed in again (1). 2. (a) jpeg (accept png) (1) / (b) pdf (1) / (c) csv (1) / (d) gif (1). 3. A midi file stores instructions / musical data such as notes, instrument and timing (1) / an mp3 stores recorded sound in compressed form (1). Accept midi files are much smaller.

System software and modes of processing

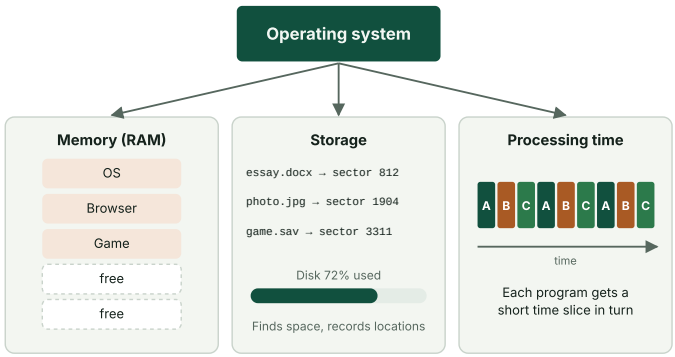
L06
FACT FILE

Specification: students should be able to

- Distinguish between system software and application software
- Describe the functions of system software, referring to allocating memory, storage and processing time
- Describe the boot-up sequence
- Describe the following modes of processing: real-time, batch and multi-user
- Choose and justify a mode of processing for a given application

CCEA fact file: Software, Software 2

Key words: software, system software, application software, operating system, memory management, processing time, time slice, boot-up sequence, real-time processing, batch processing, multi-user system, round robin, master file, transaction file, packaged software, tailored software, device driver



The operating system allocating memory, storage and processing time

System software and application software

Software is the set of instructions that tells the hardware what to do. It is split into two types. System software runs the computer itself. It acts as the link between the hardware and the programs the user wants to run. The most important piece of system software is the operating system, such as Windows, macOS, Android or iOS, which manages every other program. Utility programs and device drivers are also system software. Application software lets the user carry out a particular task, such as writing an essay, editing a photo or browsing the web. Applications can be packaged (off-the-shelf), which means they are made for many users and are cheap and ready to use, or tailored (bespoke), which means they are written for one organisation. Tailored software fits the organisation's needs exactly, but it costs more and takes time to develop, because one customer pays for all the design, programming and testing.

How the operating system manages resources

The operating system shares out the computer's resources. First, it allocates memory. A program or file must be copied from storage into RAM before the processor can use it, so the OS finds free space in RAM, gives it to the program and keeps programs apart so they do not overwrite each other. When the program closes, the space is freed. Second, it allocates storage. The OS saves files on the storage drive, records where each one is, frees the space when files are deleted and controls who can open them. Third, it allocates processing time. The processor can only do one thing at a time on each core, so the OS splits its time into tiny time slices and hands them to each task in turn. Tasks with a higher priority receive more slices. The switching is so quick that several programs appear to run at once. The OS also starts the computer: when it is switched on, instructions held in ROM check the hardware and load the operating system into RAM.

Real-time and batch processing

In real-time processing, data is processed as soon as it is entered and the output is ready quickly enough to affect the next input. Booking systems need this. When someone buys a ticket for a concert, the seat must be marked as sold straight away, or another person could buy the same seat. Air traffic control is even more critical, because an aircraft's position on screen must always be up to date. In batch processing, data is collected over a period, such as a week or a month, and then processed in one go without people being involved. It suits large amounts of similar data where an instant result is not needed. An electricity company keeps a master file of customer details. New meter readings go into a temporary transaction file. Overnight, each reading is matched to its customer using the account number, the bill is calculated, all the bills are produced together, and the master file is updated for next time.

Multi-user systems

A multi-user system lets many people, each at a different computer or terminal, share the processor of one powerful computer at the same time. It is also called a time-sharing system. The operating system gives each user a very short time slice of processor time in turn. When it has served the last user it goes back to the first, which is known as the round robin method. Because the processor works so quickly, each person feels as if they have the whole computer to themselves. For example, if four users each get a 20-millisecond slice, every user gets a turn every 80 milliseconds, which is twelve and a half times every second. Multi-user is not the same as real-time. Real-time describes how quickly data is processed, while multi-user describes how one processor is shared between many people. A system can be both, such as an online booking system used by many customers at once.

Key terms from the mark schemes

System software
Software that runs and manages the computer hardware and acts as the interface between the hardware and application programs, e.g. the operating system, utilities and device drivers.

Application software
Software designed to let the user carry out a specific task, such as word processing, spreadsheets or web browsing.

Operating system
System software that manages the computer's resources – allocating memory, storage and processing time – and lets the hardware and application software work together.
Credited: U1 2024 Q10, 2025 Q4

Boot-up sequence
The steps carried out when a computer is switched on: instructions in ROM check the hardware and load the operating system from storage into RAM.
Credited: U1 2025 Q4

Real-time processing
Each input is processed the moment it is entered, so the output is produced in time to affect the next input, e.g. ticket booking or air traffic control.
Credited: U1 2022 Q1, 2023 Q5

Batch processing
Similar data is collected into batches over a period and processed together later, without human involvement, often overnight, e.g. bills and payroll.
Credited: U1 2023 Q12

Multi-user system
A system in which many users at different terminals share the processor of one powerful computer, each given a time slice in turn (round robin).
Credited: time-sharing system; U1 2023 Q5

Time slice
A short period of processor time given to a task or user before the processor moves on to the next one.
Credited: U1 2019 Q5, 2023 Q5

Master file
A permanent or semi-permanent file holding main records (e.g. customer details and reading history) that is updated when batch processing takes place.

Transaction file
A temporary file holding the transactions collected for a batch (e.g. meter readings); it is used to update the master file.

Packaged software
Off-the-shelf software made for many users with common needs; low cost and ready to use, e.g. Microsoft Word.

Tailored software
Bespoke software developed for one organisation's specific needs; usually expensive because one customer bears the whole cost.

Exam focus

Questions on this lesson often use command words such as describe and distinguish, and they nearly always want an example. Real-time processing appeared in 2022 Q1 and was compared with multi-user processing in 2023 Q5(c). Strong answers said real-time processes each input immediately so the output affects the next input, and that multi-user shares one processor between many users using time slices. 2023 Q5(b) asked which resources the operating system manages: name memory, storage and processing time, and say what the OS does with each. Batch processing came up in 2023 Q12(a); mention collecting data over time, processing it together without human input, and a suitable example such as bills or payroll. In 2025 Q4 the Chief Examiner reported that many pupils could not describe the boot-up sequence, so learn the order: ROM instructions run, hardware is checked, the OS is loaded into RAM. Key-term matching (2024 Q10) also tested definitions from this lesson.

Unit 1 2025 Q4	[7]	Boot process, OS roles, storage conversion
Unit 1 2024 Q10	[8]	Key term matching
Unit 1 2023 Q12	[4]	Batch processing, backups, freeing disk space
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup

? Check yourself

1. Describe the difference between system software and application software. [2]
2. Classify each software example as either system software (S) or application software (A): Windows 11 operating system; Microsoft Excel spreadsheet; patient record database; disk defragmenter software; device driver; Google Chrome web browser. [6]
3. Describe two differences between packaged software and tailored software. [4]

Answers: 1. System software runs / manages the computer hardware, e.g. the operating system (1) / application software lets the user carry out a specific task, e.g. word processing (1). 2. Windows 11 – S (1) / Excel – A (1) / patient record database – A (1) / disk defragmenter – S (1) / device driver – S (1) / Google Chrome – A (1). 3. Any two differences, 2 marks each (point + contrast): packaged is made for many users, tailored for one organisation's needs (2) / packaged is cheaper because cost is shared, tailored is expensive (2) / packaged is available immediately, tailored takes time to develop (2) / packaged has lots of support / tutorials, tailored relies on the developer (2).

Utility applications and antivirus software

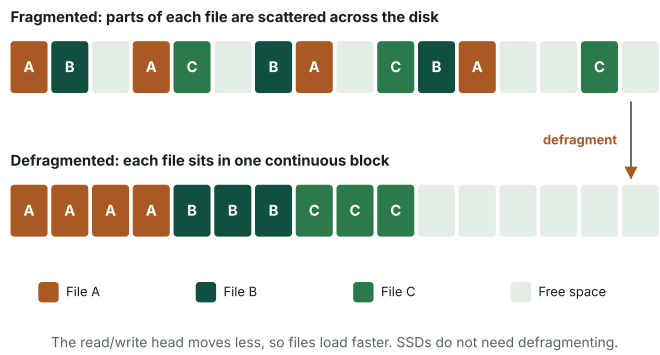
L07
FACT FILE

Specification: students should be able to

- Describe the following tasks carried out by the utility applications: disk defragmenting, task scheduling, backup and restoring data
- Explain why file fragmentation happens and how defragmentation improves performance
- Describe full and incremental backups and a suitable backup procedure
- Describe the role of antivirus software and the importance of regular updates

CCEA fact file: Software

Key words: utility software, data block, file fragmentation, disk defragmenter, read/write head, task scheduling, round robin, backup, restore, full backup, incremental backup, off-site storage, virus, antivirus software, virus signature, quarantine, update



Disk blocks before and after defragmentation

Utility software and defragmentation

Utility software is a group of small programs that each carry out one task to help the operating system keep the computer running well. They include the disk defragmenter, the task scheduler, backup and restore tools and antivirus software. To understand defragmentation, think about how files are saved. The operating system splits a file into blocks and puts each block in any free space on the disk. As files are saved and deleted over months, the free space ends up as lots of small gaps spread around the disk. A large new file may then be split into pieces stored in different places. This is called file fragmentation. On a hard disk drive, the read/write head has to keep jumping around to collect every piece, which slows the computer down. A defragmenter moves the pieces so each file is stored in one continuous area and the free space is grouped together. Solid-state drives have no moving parts, so they gain nothing from this.

Task scheduling

Task scheduling means dividing the processor's time between several tasks so the system is used as efficiently as possible. In a multi-user system, the scheduler uses the round robin method. The processor works on a task from the first computer for a short time slice, then moves to the second, then the third, and after the last one it goes back to the start. Because each slice is tiny and the processor is very fast, everybody feels that they have the computer to themselves. The word task scheduler is also used for the utility that runs jobs automatically at a planned time or when something happens. For example, a school's network manager might schedule a full virus scan every Friday evening, a backup at two in the morning and updates to install when computers shut down. The jobs run when nobody is using the system, so work is not interrupted, and they are never forgotten.

Backup and restore

A backup is a copy of data stored separately from the original so that it can be used if the original is lost or damaged. Data can be lost when a hard disk fails, when a virus or ransomware attacks, when someone deletes a file by mistake, or when a fire or flood destroys a building. For a hospital, a bank or a school, losing records could be disastrous. Backups should be kept off-site or in the cloud, somewhere secure, so a single disaster cannot destroy both copies. A full backup copies all the data, which takes a long time and a lot of space. An incremental backup only copies data that has changed since the last backup, so it is much quicker. Many organisations run a full backup once a week and incremental backups on the other nights, scheduled overnight when the network is quiet. Restoring means copying the backed-up data back onto the system to return it to its earlier state.

Antivirus software

A virus is a harmful program that attaches itself to a file. When the file is opened, the virus runs and spreads to other files and to other computers on a network or the internet. Viruses can damage or delete data, stop a computer from starting up and slow it down. Antivirus software protects against them. It scans the files on the computer's internal drive and on any connected storage, as well as data arriving through emails and downloads. It compares what it finds against a database of known virus signatures, which are patterns of code belonging to each virus. Many programs also look for suspicious behaviour that might come from a brand-new virus. When it finds a threat, the software removes it, repairs the file or moves it into quarantine so it cannot run. Because new viruses appear all the time, the signature database must be updated regularly, otherwise the newest threats will slip through undetected.

Key terms from the mark schemes

Utility software
A program that carries out one specific task to help the operating system maintain, protect or speed up the computer, e.g. defragmenter, task scheduler, backup, antivirus.
Credited: U1 2023 Q5

File fragmentation
When a file's data blocks are stored in separate, non-adjacent places on a disk because there is not enough continuous free space.

Disk defragmenter
A utility that rearranges the fragments of files on a disk so each file is stored in continuous blocks and free space is grouped together, speeding up file access.
Credited: U1 2019 Q5, 2022 Q5

Task scheduler
A utility that shares processor time between tasks (e.g. round robin) and runs jobs such as backups, virus scans or updates automatically at set times.
Credited: U1 2019 Q5, 2022 Q5

Round robin
A scheduling method in which each task or user is given a time slice of the processor in turn, returning to the first after the last.

Backup
A copy of data stored separately, preferably off-site, so it can be restored if the original is lost or damaged.
Credited: U1 2022 Q1, 2023 Q12, 2024 Q10

Restore
Copying data back from a backup to return data or a system to an earlier state.
Credited: U1 2022 Q5

Full backup
A backup in which all the data is copied; it takes a long time and needs a lot of storage.

Incremental backup
A backup that copies only the data that has changed since the last backup; quicker and smaller than a full backup.

Virus
A harmful program that attaches itself to a file and spreads to other files and computers when the file is opened.

Antivirus software
Utility software that detects viruses by comparing files with a database of virus signatures, then removes or quarantines them; it must be updated regularly.
Credited: U1 2022 Q5

Quarantine
Isolating an infected file so it cannot run or spread, while keeping it in case it needs to be restored.

Exam focus

Utilities are tested often. In 2019 Q5(b) pupils had to describe what disk defragmenting and task scheduling do. For defragmenting, the marks come from three ideas: files become split into fragments across the disk, the defragmenter moves the fragments so each file is stored together, and this makes files faster to access. Writing only 'it tidies up the disk' will not score. For task scheduling, say that it shares processor time between tasks or that it runs jobs such as backups or virus scans automatically at set times. Utilities and security came up in 2022 Q5(b) and 2023 Q5(d). Backups appeared in 2022 Q1 and 2023 Q12(b), and freeing up disk space in 2023 Q12(c). Key-term matching in 2024 Q10 included backup. When explaining why antivirus must be updated, link the two ideas: new viruses keep appearing, and the software can only detect viruses whose signatures are in its database.

Unit 1 2024 Q10	[8]	Key term matching
Unit 1 2023 Q12	[4]	Batch processing, backups, freeing disk space
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q5	[6]	3D printing, utilities and security, graphics digitiser
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup

? Check yourself

1. The operating system on John's computer includes a utility application. What is the function of the following two tasks carried out by utility software? (i) Disk defragmenting (ii) Task scheduling [4]
2. Explain why files become fragmented on a hard disk drive. [3]
3. Describe how disk defragmenting improves the performance of a computer. [3]

Answers: 1. (i) Defragmenting: files become split into fragments stored in different places on the disk (1) / the defragmenter rearranges the fragments so each file is stored in continuous blocks / free space grouped together (1). (ii) Task scheduling: divides processor time between tasks / users, e.g. round robin (1) / or runs jobs such as backups / virus scans automatically at set times (1). 2. The OS breaks files into blocks and stores them in any free space (1) / as files are saved and deleted, free space is left in small gaps spread across the disk (1) / so a new large file has to be split between several non-adjacent gaps (1). 3. Moves the blocks of each file so they are stored together / contiguously (1) / groups free space together (1) / the read/write head moves less, so files are read / saved more quickly (1).

Database concepts

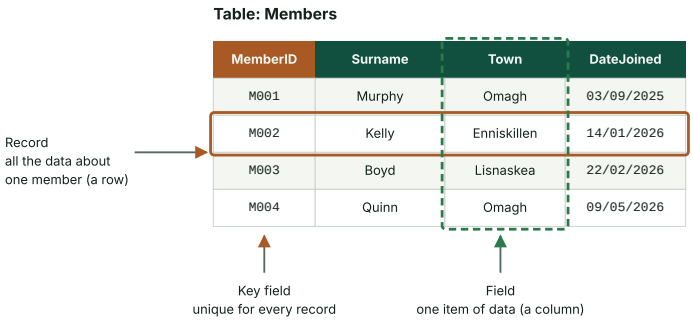
L08
FACT FILE

Specification: students should be able to

- Demonstrate understanding of and explain basic database concepts such as table, record, field, key field, query, form, report, macro, relationship and importing data
- Explain the difference between a flat file and a relational database
- Explain how a primary key and a foreign key are used to create a relationship between tables
- Describe one-to-one, one-to-many and many-to-many relationships and the purpose of referential integrity
- Describe the purpose of queries, forms, reports and macros in a database application

CCEA fact file: Fact File 07: Database Applications (1)

Key words: database, table, flat file, record, field, key field, primary key, composite key, foreign key, relationship, one-to-many, referential integrity, query, form, report, macro, importing, data redundancy



A database table showing a record, a field and the key field

What a database is and why we use one

A database is an organised collection of related data, stored so that it can be searched, sorted and presented quickly. Almost every organisation in Northern Ireland uses one: schools keep pupil details and attendance, GP surgeries keep patient records, shops keep stock levels and sports clubs keep member lists. Inside the database, data is held in tables. A table stores data about one type of thing, for example members, and is laid out in rows and columns. A database package also gives tools to work with the data. Queries find the records you need, forms make it easy to enter and edit data, reports present the results neatly, and macros automate jobs that are done again and again. Tables can be linked by relationships so that each fact only needs to be stored once. In this lesson you will learn the key terms that the exam expects you to define and to spot in a table.

Records, fields and key fields

Each row in a table is a record: all of the data about one person or item, such as one swimmer in a club. Each column is a field: a single item of data that is stored for every record, such as Surname or DateOfBirth. The column heading is the field name, and each field is given a data type. The key field, also called the primary key, is the field that uniquely identifies each record. No two records can have the same key value and it cannot be left blank. Codes such as MemberID or PupilID make good key fields because the organisation creates them to be unique. Names, dates of birth, phone numbers and towns make poor key fields because two people can share them. Sometimes no single field is unique, so two fields are used together; this is called a composite key, for example MemberID and SessionDate in a table of lesson attendance.

Flat files, relationships and referential integrity

A flat file keeps all the data in one table. This often means the same details are typed many times, which is called data redundancy. It wastes space and causes errors, because one copy may be updated while another is not. A relational database avoids this by splitting the data into several tables, each about one thing, and linking them. The link is made by storing the primary key of one table in the other table, where it is called a foreign key. For example, MemberID is the primary key of Members and a foreign key in Lessons. This is a one-to-many relationship: one member can book many lessons. Other types are one-to-one and many-to-many. Enforcing referential integrity on the relationship keeps the tables consistent. It stops a lesson being booked for a member who does not exist, and stops a member being deleted while bookings still point to them, unless cascade delete is switched on.

Queries, forms and reports

Tables store the data, but users rarely work with tables directly. A query is a question asked of the database. It checks every record and returns only those that meet the criteria, for example all Junior squad members born after 2011. A query can use fields from more than one linked table and can sort the results. A form is an on-screen layout used to enter new records, edit existing ones or view records, usually one at a time. Forms are easier to use than a table, can include drop-down lists, buttons and the club logo, and can combine fields from more than one table while saving each item to the correct table. They also help keep data accurate by limiting what can be typed. A report presents data from a table or query in a formatted layout ready for printing or display. Reports can group similar records, sort them and calculate totals or counts, for example a register of members grouped by squad.

Macros and importing data

A macro is a small stored program that performs a set of database actions automatically. Instead of carrying out several steps every time, the user clicks one button on a form and the macro does the work. Macros can open a form, run a query, print a report, or add, update or delete a record. These four record jobs are sometimes called CRUD: create, read, update and delete. Macros save time, reduce mistakes and let staff with little database knowledge use the system safely. Importing data means bringing data into the database from another file rather than typing it again. A common example is importing a csv file or a spreadsheet worksheet, such as a list of new members exported from an online sign-up form. Importing is quicker and avoids typing errors. Before importing, the data should be checked so that column headings match the field names, values suit each field's data type and there are no duplicate key values.

T Key terms from the mark schemes

Database
An organised, structured collection of related data that can be searched, sorted and presented, usually held in one or more tables.

Flat file
A database that holds all its data in a single two-dimensional table, which often leads to repeated data.

Field
A single item of data stored about every object or person, forming a column in a table and identified by its field name.
`Credited: U1 2019 Q3, 2021 Q2, 2024 Q3`

Table
A database structure that stores data about one type of thing, organised in rows (records) and columns (fields).
`Credited: U1 2019 Q3, 2024 Q3`

Record
All of the related data about one object or person, forming a complete row in a table. A record is made up of fields.
`Credited: U1 2019 Q3, 2023 Q3, 2024 Q3`

Key field / primary key
A field whose value is unique for every record, used to identify each record in a table.
`Credited: U1 2019 Q3, 2022 Q10, 2024 Q3`

Composite key
A primary key made up of two or more fields that together uniquely identify a record.

Foreign key
The primary key of one table stored as a field in a second table, used to link the two tables.

Relationship
A link between two tables made through a common field (primary key and foreign key). It can be one-to-one, one-to-many or many-to-many.

Relational database
A database in which data is held in several linked tables.
`Credited: U1 2021 Q2, 2024 Q3`

One-to-many relationship
One record in a table links to many records in another table, e.g. one customer makes many bookings.
`Credited: U1 2019 Q3, 2021 Q2`

Cardinality
How the records in two related tables relate to each other: one-to-one, one-to-many or many-to-many.

Crow's foot
A symbol drawn at the 'many' end of a relationship line, used instead of the infinity sign (∞).

Referential integrity
A rule applied to a relationship that keeps the linked tables consistent, e.g. a foreign key value must match an existing primary key.

Form
An on-screen layout used to enter new records, edit existing records or view records in a database.
`Credited: U1 2022 Q10, 2023 Q3`

Report
A database tool that presents data from a table or query in a formatted, user-friendly layout for printing or display, often grouped and sorted.
`Credited: U1 2019 Q3, 2022 Q10, 2023 Q3`

Macro
A small stored program that carries out a repetitive task automatically, usually at the click of a button.
`Credited: U1 2021 Q3, 2022 Q10, 2024 Q10`

Import
To bring data into an application from another file, such as a csv file or spreadsheet, instead of retyping it.
`Credited: U1 2022 Q10`

Exam focus

Database questions appear on nearly every Unit 1 paper, often worth 8 to 11 marks: 2019 Q3, 2021 Q2, 2022 Q10, 2023 Q3, 2024 Q3 and 2025 Q5. You are often given a table and asked to name a record, a field or the key field, or to fill in missing field names. When asked to choose a key field, name it and justify it: the value is unique for each record. Say why other fields are unsuitable, for example two customers could have the same surname. For relationship questions (2019 Q3, 2021 Q2) name the type, usually one-to-many, and say which table is the 'one' end. Questions on database tools (2022 Q10, 2023 Q3) and key term matching (2024 Q10) test whether you can tell a query, form, report and macro apart, so learn a clear one-sentence definition of each. A common mistake is saying a report searches for data; it is the query that searches and the report that presents the results.

`Unit 1 2025 Q5`	`[11]`	Data types, validation checks, data processing
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q3`	`[11]`	Field names and data types, database terms, key field, query design
`Unit 1 2023 Q3`	`[9]`	Database concepts; validation rules
`Unit 1 2022 Q10`	`[11]`	Database tools, field types, importing csv, validation

? Check yourself

1. Define the terms (a) record and (b) field. [2]
2. Look at the Erne Valley Swimming Club Members table on the slides. (a) State how many records the table contains. (b) Identify the key field and justify your choice. (c) Explain why Surname would not be a suitable key field. [4]
3. Describe the purpose of a form in a database, and give two ways a form can help to keep the data accurate. [3]

Answers: 1. (a) All the data about one person or item / one row in a table (1). (b) One item of data stored about each record / one column in a table (1). 2. (a) 4 (1). (b) MemberID (1) because each member has a different/unique ID (1). (c) Two members can have the same surname, e.g. Maguire (1) so it would not uniquely identify a record (1). 3. Purpose: an on-screen layout used to enter new records / edit or view existing records (1). Any two from: drop-down lists limit choices (1) / validation rules on fields reject unsuitable data (1) / input masks control format (1) / only the fields needed are shown so fewer mistakes (1) / data is saved to the correct table automatically (1).

Database data types and validation

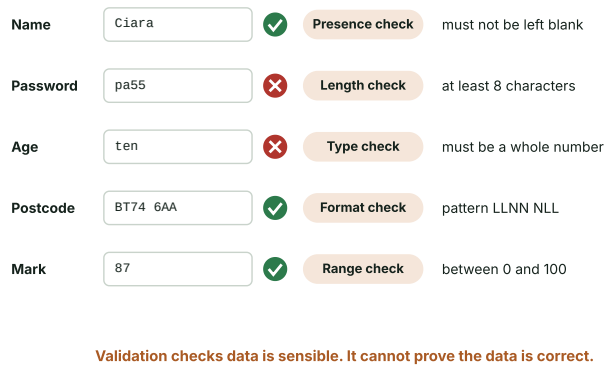
L09 FACT FILE

Specification: students should be able to

- Identify and use appropriate data types when creating a database structure
- Justify the choice of data type for a field
- Demonstrate understanding of the need for data validation
- Describe the following types of validation checks: presence, length, type, format and range
- Write simple validation rules for database fields

CCEA fact file: Fact File 08: Database Applications (2)

Key words: data type, Short Text, Long Text, Number, Currency, Date/Time, AutoNumber, Yes/No, OLE Object, Hyperlink, Attachment, Calculated, Lookup Wizard, validation, presence check, length check, type check, format check, range check, input mask, validation rule, validation text, verification



Five validation checks applied to form fields, with passes and failures

Data types

When a database table is designed, every field is given a field name and a data type. The data type decides what kind of data the field can hold and what can be done with it. Short Text is for names, addresses, postcodes and any numbers that are not used in calculations, such as phone numbers. Long Text holds longer descriptions. Number is for numeric values that will be used in calculations, such as the number of people in a group. Currency is for money and shows two decimal places. Date/Time stores dates and times so that they can be sorted in date order and used to work out ages or due dates. AutoNumber creates a new number automatically for each record and is often used for the key field. Yes/No holds a Boolean value with only two options. Choosing the right data type is the first step in keeping data accurate, because the database will refuse data of the wrong type.

More data types and justifying your choice

Database packages such as Access offer several more data types. Hyperlink stores a web address or email address that can be clicked. Attachment lets files, such as a photo of a club member, be attached to a record. OLE Object, short for Object Linking and Embedding, stores a file created in another program, such as a spreadsheet or a word-processed document. Calculated holds the result of a calculation that uses other fields in the same record, for example a total cost worked out from the number in the group multiplied by the price per person. Lookup Wizard creates a drop-down list of values, so the user picks an answer rather than typing it. In the exam you may need to justify a data type. Give a reason linked to the data. For example, a phone number should be Short Text because it starts with a zero, may contain spaces and is never used in a calculation. Storing it as a Number would drop the leading zero.

Why validation is needed

Validation is an automatic check carried out by the computer when data is entered. It makes sure that the data is sensible, reasonable and within acceptable limits. If the data breaks a rule, it is rejected and an error message, called validation text, tells the user what is wrong. Validation matters because a database is only useful if its data is accurate. Wrong or missing data leads to wrong results, for example a letter sent to an incomplete address or a bill calculated from an impossible number of days. Validation reduces these input errors. However, validation cannot check that data is correct. It can only check that data is allowed. If a parent's date of birth is typed as 12/05/1984 instead of 21/05/1984, both dates are valid, so the mistake gets through. This is why validation is often combined with verification, such as typing an email address twice, or a visual check against the original document.

The five validation checks

A presence check makes sure a field is not left blank. It suits fields that every record must have, such as a customer's surname or postcode. A length check makes sure the data has the right number of characters. It can set an exact length, such as a seven-character camper code, or a minimum or maximum, such as a password of at least eight characters. A type check makes sure the data is of the correct data type, so letters cannot be typed into a Number or Date/Time field. A format check makes sure the data matches a set pattern, such as a Northern Ireland postcode that starts with BT followed by digits, a space, a digit and two letters. A range check makes sure a value lies between a lower limit and an upper limit, for example from 1 to 5 days at a summer camp or a test mark from 0 to 100. When answering, always name the check and then describe it using the field in the question.

Writing validation rules in Access

In Access, most checks are set in the field properties in Design View. For a presence check, set Required to Yes, or enter the validation rule Is Not Null. For a length check, set the Field Size, or write a rule using the Len function, such as Len([Password])>=8, which counts the characters in the field. A type check happens automatically once the data type is chosen. For a format check, use an input mask. In a mask, L means a letter must be typed, 0 means a digit must be typed and 9 means a digit is optional, so the mask "BT"09 0LL fits Northern Ireland postcodes. For a range check, write a rule with comparison operators, such as >=1 And <=5, or use Between 1 And 5, which includes both end values. Always add validation text that explains the rule in plain English, so the user knows how to fix the error rather than just seeing that it was rejected.

T Key terms from the mark schemes

Data type
The kind of data a field can hold, e.g. Short Text, Number, Currency or Date/Time. It controls what can be entered and what can be done with the data.
Credited: U1 2019 Q3, 2024 Q3, 2025 Q5

Validation
An automatic check made by the computer as data is input, to make sure the data is sensible, reasonable and inside allowed limits. Data that breaks the rules is rejected; validation cannot confirm the data is correct.
Credited: U1 2021 Q3, 2025 Q5; U4 2025 Q6

Presence check
A check that makes sure data has been entered in a required field, so the field cannot be left blank.
Credited: U1 2021 Q2, 2022 Q10, 2025 Q5

Length check
A check that makes sure the data entered has an exact number of characters, or is no shorter or longer than set limits.
Credited: U1 2022 Q10, 2025 Q5

Type check
A check that makes sure the data entered is of the correct data type, e.g. numeric, text or date.

Format check
A check that makes sure the data entered matches a set pattern, e.g. a postcode or an email address containing @.
Credited: U1 2023 Q3, 2025 Q5

Range check
A check that makes sure the data entered lies between a lower limit and an upper limit.
Credited: U1 2021 Q2, 2022 Q10, 2025 Q5

Verification
A check that data has been entered or copied accurately, e.g. by double entry or a visual check against the source.

Currency
A data type for money values, stored and displayed with two decimal places.

OLE Object
Object Linking and Embedding: a data type for a file or object created by software other than the database, e.g. a spreadsheet or word-processed document.

Hyperlink
A data type that stores a link to a web address (URL) or an email address.

Lookup Wizard
Creates a list of values so that a value can be chosen from a drop-down list and entered into the field.

Input mask
A pattern set on a field that controls which characters can be typed in each position, used to carry out a format check.

AutoNumber
A data type, often used for a key field, in which the computer creates a new sequential number automatically for each record.

Yes/No
A data type that holds one of only two values (a Boolean), such as Yes/No or True/False.

Attachment
A data type that lets files, such as a digital image, be attached to a record.

Calculated field
A data type that stores the result of a calculation using other fields in the same record.

Exam focus

Data types and validation have come up often: 2019 Q3 and 2024 Q3 asked for data types with a justification, 2022 Q10 covered field types and validation, 2021 Q2(g) asked about presence and range checks, 2023 Q3 asked for validation rules, and 2025 Q5 covered data types and validation checks. Most marks come in pairs: one mark for naming the check or data type and one for a reason or description in context. 'Range check' alone will rarely get both marks; 'a range check so that the number of days booked must be from 1 to 5' will. In 2025 many candidates did not realise that validation cannot tell whether data is correct, so learn that point. When writing a rule, check the boundaries carefully: for 5 to 20 people, >=5 And <=20 is right, while >5 And <20 wrongly rejects 5 and 20. Do not confuse a length check, which counts characters, with a range check, which compares values.

Unit 1 2025 Q5	[11]	Data types, validation checks, data processing
Unit 1 2023 Q3	[9]	Database concepts; validation rules
Unit 1 2022 Q10	[11]	Database tools, field types, importing csv, validation
Unit 1 2021 Q2(g)	[3]	Presence and range checks

? Check yourself

1. Why is validation used when data is entered into a database? [2]
2. Explain why validation cannot guarantee that data is correct. Use an example. [2]
3. Describe a range check and give an example of a field where one would be used. [2]

Answers: 1. Any two from: to check data is sensible/reasonable/within acceptable limits (1) / to reduce input errors (1) / so that invalid data is rejected at the input stage (1) / so the database is more accurate/reliable (1). 2. Validation only checks that data is allowed/follows the rules, not that it is the true value (1). Example, e.g. a date of birth typed as 12/05/1984 instead of 21/05/1984 is still a valid date and is accepted (1). 3. Checks that a value lies between a lower and an upper limit (1). Example, e.g. an exam mark must be from 0 to 100 / age 11 to 19 (1).

Queries, logical operators and big data

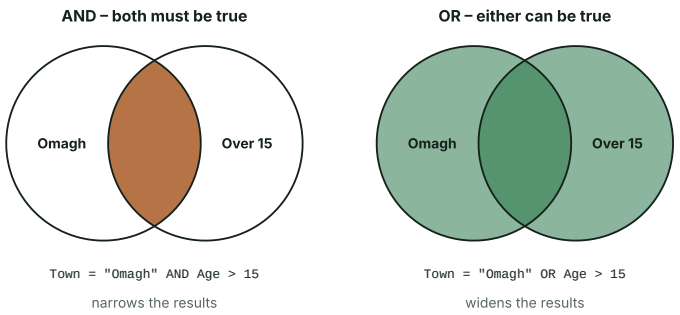
L10 FACT FILE

Specification: students should be able to

- Extract data from a database structure using simple query structures and using the following logical operators: <, >, =, <=, >=, AND, OR and BETWEEN
- Predict the records returned by a query with given criteria
- Demonstrate understanding of big data, referring to volume, velocity and variety
- Describe the difference between structured and unstructured data
- Demonstrate understanding of the need for data analytics to interpret big data

CCEA fact file: Fact File 08: Database Applications (2)

Key words: query, criteria, query design grid, comparison operator, logical operator, Boolean, AND, OR, BETWEEN, <>, big data, volume, velocity, variety, structured data, unstructured data, data analytics, data mining



`Town = "Omagh" AND Age > 15`

narrows the results

`Town = "Omagh" OR Age > 15`

widens the results

BETWEEN 10 AND 20 includes both 10 and 20.

Venn diagrams: AND returns records in both sets, OR returns records in either

Queries and comparison operators

A query is a question asked of a database. It checks every record and returns only those that meet the criteria, which are the conditions typed into the query. Most criteria use comparison operators. The equals sign finds an exact match, such as Club = "Lisnaskea". Less than (<) and greater than (>) exclude the value itself, so Goals > 8 does not return a player with exactly 8 goals. Less than or equal to (<=) and greater than or equal to (>=) include the value, so Score >= 50 returns a score of 50. Not equal to (<>) returns everything except the value given. Each comparison gives a Boolean result: the record either meets the condition (true) or does not (false). When writing criteria, put text values in quotation marks, and type them exactly as they are stored in the table. In Access, dates are written between hash signs, such as #01/03/2026#, so that they are treated as dates rather than text.

AND, OR and BETWEEN

Logical operators join two or more conditions together. With AND, both conditions must be true for the record to be returned, so AND usually reduces the number of results. For example, Club = "Lisnaskea" AND Goals > 10 only finds Lisnaskea players who have also scored more than ten goals. With OR, a record is returned if at least one condition is true, so OR usually increases the number of results. Club = "Lisnaskea" OR Club = "Irvinestown" finds players from either club. BETWEEN finds values within a range, and the range includes both end values. Age BETWEEN 13 AND 14 returns players aged 13 and 14, and gives the same results as Age >= 13 AND Age <= 14. A common mistake is to use AND when OR is needed: Club = "Lisnaskea" AND Club = "Irvinestown" returns nothing, because one player cannot belong to both clubs at the same time.

Building a query in design view

In Access, queries are usually built in the query design grid. First add the table or linked tables you need. Then place each field you want in the Field row. The Table row shows which table the field comes from. The Sort row puts the results in ascending or descending order, and the Show tick box decides whether a field appears in the results. A field can be used for criteria but hidden, for example to search by club without showing the club name. Criteria are typed in the Criteria row under the field they apply to. Conditions typed on the same row are joined with AND. Conditions typed on different rows, using the Criteria row and the Or row, are joined with OR. Once the query is run, its results can be used as the source of a report. This is how a club could print a list of players for one team, sorted by surname, without changing the data in the table.

Big data and the three Vs

Big data means data sets that are so large, arrive so fast and are so mixed that ordinary database software cannot store or process them in a sensible time. It is usually described using three Vs. Volume is the huge amount of data, often measured in terabytes or petabytes, such as every loyalty-card purchase made in UK supermarkets. Velocity is the speed at which new data is created and must be processed, often in real time. A bank must check a card payment for fraud within a second, and a social media post should appear straight away. Variety is the many different types and formats of data, such as text, photos, video, GPS locations and readings from sensors. Some of this is structured data, organised in tables with fixed fields such as names and dates. Much of it is unstructured data, such as posts, emails and video, which has no fixed format and cannot easily be searched with normal database tools.

Why data analytics is needed

Collecting big data is only useful if something can be learned from it. Data analytics means using algorithms, programs or artificial intelligence to examine data and find patterns, trends and relationships. Data mining is a part of this: searching very large data sets for patterns that nobody knew were there. Analytics is needed because big data is far too large and changes too quickly for people to examine by hand, and because it is in many formats that ordinary queries cannot handle. It needs specialist software, powerful computers, often hired from cloud providers, and skilled analysts. Examples include a supermarket using sales data to decide which offers to send each customer, a streaming service recommending films, a bank spotting unusual spending that may be fraud, and a health trust predicting busy periods in emergency departments so that more staff are on duty. Analytics helps organisations make decisions based on evidence rather than guesswork.

T Key terms from the mark schemes

Query
A way of searching a database to extract the records that meet set criteria. It checks each record and lists the data that satisfies the criteria.
Credited: U1 2021 Q2, 2023 Q3

Criteria
The conditions a record must meet to be included in the results of a query, e.g. Age >= 18.

Comparison operator
A symbol used to compare a value with other data, giving true or false: < less than, <= less than or equal to, > greater than, >= greater than or equal to, = equal to, <> not equal to.

Logical operator
An operator such as AND, OR or BETWEEN that combines or tests conditions and gives a true or false (Boolean) result.

BETWEEN
A logical operator that selects values within a range, including both end values.

Group by
Organises query or report results by the values of a field, e.g. grouping members by squad.
Credited: U1 2021 Q2

Big data
Very large sets of data that could be mined for information but cannot be analysed by normal means because of their volume, velocity and variety.
Credited: U1 2021 Q8, 2022 Q1

Volume
The very large amount of data stored by organisations, e.g. terabytes of social media posts each day.

Velocity
The speed at which data is created and must be processed, often in real time.

Variety
The many different types and formats of data, such as text, images, video and sensor readings, which may be structured or unstructured.

Structured data
Data organised in a fixed format, usually tables of fields and records, so it is easy to search and analyse, e.g. dates, names, addresses, card numbers.

Unstructured data
Data with no fixed format that cannot easily be processed with normal database tools, e.g. social media posts, emails, videos and essays.

Data analytics
Using algorithms, programs or artificial intelligence to examine data and find patterns, trends and relationships so that conclusions can be drawn.

Data mining
Searching very large data sets with software to discover patterns and relationships that were not previously known.

Exam focus

Query questions usually give you a table and ask you to complete criteria in a query design grid, or to say which records a query returns: 2019 Q3(d), 2021 Q2(f) and 2024 Q3(e). Copy field names and values exactly as they appear in the table, put text in quotation marks, and check boundary values carefully: > 15 does not include 15. Remember that criteria on the same row mean AND, while criteria on separate rows mean OR. Big data has appeared as a three-mark question on volume, velocity and variety (2021 Q8(b)) and in the multiple-choice question in 2022 Q1. For each V, give a short definition and an example: one mark is often given for each V explained. Do not mix them up. Velocity is about speed, not size, and variety is about different types of data, not different amounts. If asked why analytics is needed, explain that big data is too large and complex to analyse by normal means and that its value comes from the patterns found.

Unit 1 2024 Q3	[11]	Field names and data types, database terms, key field, query design
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup
Unit 1 2021 Q8(b)	[3]	Big data: volume, velocity, variety
Unit 1 2021 Q2(d)–(f)	[5]	Export to csv, queries, grouping
Unit 1 2019 Q3(d)–(f)	[5]	Query criteria, reports, sorting, relationship type

? Check yourself

1. Write the criteria to find all members who are aged 18 or over. [1]
2. Write the criteria to find all bookings made between 1 March 2026 and 31 March 2026. [2]
3. Explain the difference between the AND and OR operators in a query. [2]

Answers: 1. >=18 (in the Age field) (1). Accept Age >= 18 or >17. 2. Between #01/03/2026# And #31/03/2026# (1 for BETWEEN or >= and <= used, 1 for both correct dates). Accept >=#01/03/2026# And <=#31/03/2026#. 3. AND: both/all conditions must be true for a record to be returned (1). OR: only one/at least one condition needs to be true (1).

Spreadsheets 1: structure, formatting, formulae and cell referencing

L11 FACT FILE

Specification: students should be able to

- Describe the basic structures of spreadsheet software: workbooks, worksheets, cells, rows, columns and ranges
- Describe and use data types; enter text, numbers and formulae; format cells, rows and columns
- Create and replicate formulae using arithmetic operators
- Use simple functions (SUM, AVERAGE, MAX, MIN, COUNT, COUNTIF) and relative and absolute cell referencing
- Improve worksheet presentation with headers and footers, and select areas of a spreadsheet for printing

CCEA fact file: Spreadsheets 1

Key words: worksheet, cell, row, column, cell reference, range, label, value, data format, formula, function, operator, replicate, fill handle, relative reference, absolute reference, merge and centre, print area



Spreadsheet formulas copied down: relative references change, the absolute reference stays fixed

How a spreadsheet is organised

A spreadsheet is application software that stores data in a grid so that it can be calculated, analysed and presented. One spreadsheet file, called a workbook, can contain several worksheets, each shown on its own tab. A worksheet is made of columns, labelled with letters, and rows, labelled with numbers. The box where a column and a row cross is a cell. Each cell has a cell reference, written column first and row second, so H8 means column H, row 8, just like co-ordinates in maths. A group of cells next to each other is a range, written with a colon between the first and last cell, for example B2:B10. Cells can hold labels (text that explains what data means), values (numbers used in calculations), dates, formulae, or a reference to another cell. The big advantage of a spreadsheet over a paper table is automatic recalculation: when one value changes, every formula that depends on it is worked out again straight away.

Data formats and presentation

Formatting changes how data looks without changing the value stored. A number can be shown as General, Number with a fixed number of decimal places, Currency (£), Percentage, Date or Time. Choosing the right format makes data easier to read and helps avoid mistakes: a cost column should be Currency, a discount Percentage, and a phone number Text so that the leading zero is kept. Percentage format shows 0.2 as 20%, so a rate typed as 20% is really stored as 0.2. Presentation features include bold or larger fonts for headings, borders, fill colour, alignment, merge and centre for a title that runs across several columns, and wrap text so long labels fit inside a cell. Column widths can be changed so that nothing shows as ####. For printing, headers and footers can add a title, file name, date or page number to every page, and a print area can be set so that only the part of the sheet that matters is printed.

Formulae and simple functions

A formula tells the spreadsheet what calculation to do. It must always begin with an equals sign, then use cell references, numbers and operators: + to add, - to subtract, * to multiply, / to divide and ^ for a power. Brackets are calculated first, so =(B2+C2)/2 finds the mean of two cells. Using cell references rather than typing the numbers means that the answer updates when the data changes. A function is a ready-made formula that does a particular job. SUM adds a range, AVERAGE finds the mean, MAX returns the largest value and MIN the smallest. COUNT counts how many cells in a range contain numbers, and COUNTIF counts only the cells that meet a condition, such as =COUNTIF(C2:C20,"Yes"). In each case the range goes inside the brackets with a colon between the first and last cell. =SUM(A1:A10) adds every cell from A1 to A10, not just the two cells named.

Replication and cell referencing

Replicating a formula means copying it into other cells, usually by dragging the fill handle down a column or across a row. This is quicker than typing each formula and makes errors less likely. By default references are relative: the spreadsheet adjusts them to fit the new position. If D2 contains =B2*C2 and it is filled down, D3 contains =B3*C3 and D9 contains =B9*C9. Sometimes every formula must point to one fixed cell, for example a single VAT rate, discount rate or exchange rate stored at the top of the sheet. This needs an absolute reference, made by putting a dollar sign in front of the column letter and the row number, such as $F$1. When =G2*(1+$F$1) is filled down, the G reference changes each row but $F$1 stays locked. If the dollar signs are missed, the copied formula points at the wrong cell, often a blank one, and gives wrong results without any warning. Keeping the rate in one cell also means it can be changed once and every result updates.

T Key terms from the mark schemes

Spreadsheet
Application software that organises data in a grid of rows and columns so it can be calculated, analysed and presented. A spreadsheet file (workbook) contains one or more worksheets.

Cell range
A block of adjacent cells referred to together, written first cell:last cell, e.g. B5:B12.
Credited: U1 2021 Q3, 2024 Q4, 2025 Q6

Label
Text used to identify the contents of a row or column, e.g. Name, Age.

Value
A number entered into a spreadsheet so that calculations can be carried out, e.g. 5, 23.

Formula
An instruction in a cell to carry out a calculation using operators and cell references, e.g. =A1+B1. It always starts with = and the cell shows the result, which updates when the referenced cells change.

Spreadsheet function
A built-in formula that does a specific job, e.g. SUM, AVERAGE, MAX, MIN, COUNT, IF and VLOOKUP.

SUM function
Adds all the numbers in a range, e.g. =SUM(A1:A10) adds A1 through to A10.
Credited: U1 2023 Q4, 2024 Q4, 2025 Q6

AVERAGE function
Calculates the arithmetic mean of the values in a range, e.g. =AVERAGE(A1:A10).

MAX function
Returns the highest value in a range, e.g. =MAX(A1:A10).

MIN function
Returns the lowest value in a range, e.g. =MIN(A1:A10).

COUNT / COUNTIF
COUNT counts the cells in a range that contain numbers; COUNTIF counts only the cells that meet a condition, e.g. =COUNTIF(C2:C20,"Yes").

Data types (spreadsheet formats)
Formats applied to cell contents to show what kind of data is stored, e.g. Currency, Number, Text, Date/Time, Percentage.

Replicating formulae
Copying a formula down or across cells (by dragging the fill handle or Fill Down) so the spreadsheet adjusts the references to match each row or column. Saves time and reduces errors.
Credited: Replication

Fill handle
The small square at the bottom right of a selected cell; dragging it replicates the cell's formula into neighbouring cells.

Relative cell reference
A cell reference that changes when the formula is replicated, so it refers to the matching cell in the new row or column, e.g. B9.
Credited: U1 2021 Q3

Absolute cell reference
A cell reference that stays the same when the formula is copied, replicated or dragged down, shown with $ signs, e.g. $B$9.
Credited: U1 2021 Q3, 2023 Q4, 2024 Q4, 2025 Q6

Merge and centre
Combining two or more adjacent cells into a single cell and centring the contents across it.
Credited: U1 2022 Q3, 2023 Q4

Wrap text
Formatting that makes long text continue onto extra lines inside the cell instead of spilling into the next cell.
Credited: U1 2024 Q4

Headers and footers
Text placed at the top (header) or bottom (footer) of every printed page, e.g. a title, file name, date or page number.

Print area
A selected range of cells that is the only part of the worksheet printed.

Exam focus

Spreadsheet questions appear on every Unit 1 paper, usually worth 10 to 13 marks and based on a screenshot of a sheet. You will often be asked to write a formula or function for a named cell: SUM with a correct range (2023 Q4, 2024 Q4(a), 2025 Q6(d)), AVERAGE or MAX (2021 Q3), or a subtraction (2024 Q4). Full marks need the equals sign, the right function name, brackets and a colon in the range, for example =SUM(B5:B12). Using x instead of * or listing cells one by one when a range was asked for loses marks. Absolute referencing is tested often (2019 Q8(c), 2021 Q3, 2023 Q4(c), 2024 Q4(e)): say that the reference does not change when the formula is replicated, and explain why it matters in that sheet, such as one exchange rate used by every row. Formatting questions ask you to name or describe a feature you can see, such as merge and centre (2022 Q3(b), 2023 Q4(a)), currency (2022 Q3(a)) or wrap text (2024 Q4). Name the feature precisely and say what it does to the cells.

Unit 1 2025 Q6	[10]	Spreadsheet features, ranges, SUM, IF with absolute reference, lookup functions
Unit 1 2024 Q4	[13]	SUM, subtraction, wrap text, absolute references, chart ranges and type
Unit 1 2023 Q4	[11]	Merge and centre, SUM, formulae, absolute references, IF, charts, template
Unit 1 2022 Q3	[12]	Currency, merge and centre, formulae, ranges, nested IF, VLOOKUP
Unit 1 2021 Q3(a)–(i)	[12]	Formatting, AVERAGE, replication, referencing, templates, validation, nested IF, MAX, ranges, macros

? Check yourself

1. Describe the difference between a row and a column in a spreadsheet. [2]
2. A sports club records match fees in cells B2 to B12. Write a function to add the values in B2 to B12. [1]
3. Write a function to find the highest value in cells C2 to C20. [1]

Answers: 1. A row runs horizontally / across the sheet and is identified by a number (1) / A column runs vertically / down the sheet and is identified by a letter (1) 2. =SUM(B2:B12) (1) – equals sign, function and correct range all needed 3. =MAX(C2:C20) (1)

Spreadsheets 2: IF, VLOOKUP, validation, templates, modelling, charts and macros

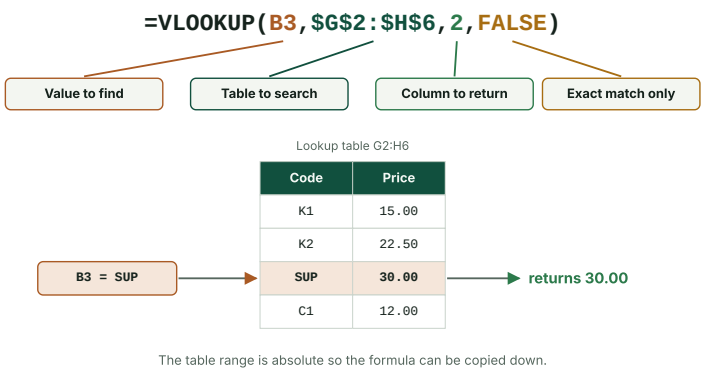
L12 FACT FILE

Specification: students should be able to

- Describe and use conditional formatting and data validation
- Describe and use templates, headers and footers, and importing data; create a simple template for others to use
- Use IF statements (including nested IF) and VLOOKUPs
- Use a spreadsheet for data modelling and what-if analysis
- Create, label and format charts
- Create simple macros

CCEA fact file: Spreadsheet 2

Key words: IF, nested IF, condition, VLOOKUP, lookup table, conditional formatting, data validation, range check, drop-down list, error alert, template, import, CSV, model, what-if, chart, legend, macro



The four parts of a VLOOKUP formula and the row it matches in a lookup table

Making spreadsheets work harder

Beyond simple sums, spreadsheets have features that make decisions, find information, check input and save repeated work. The IF function tests a condition and returns one result when it is true and another when it is false, so a sheet can show Pass or Fail, add a delivery charge, or choose between two exchange rates on its own. VLOOKUP searches the first column of a lookup table and returns matching information, such as a price, from the same row. Conditional formatting changes how a cell looks when a condition is met, so important values stand out. Data validation limits what can be typed into a cell, and an error alert asks the user to try again. Templates give a ready-made layout, and importing brings in data from other software without retyping. Spreadsheets can also be used as models to answer what-if questions, charts show results visually, and macros automate tasks that are done again and again. Each feature saves time, reduces errors or makes information easier to understand.

IF, nested IF and VLOOKUP

An IF function has three parts inside its brackets, separated by commas: the condition, the value if true and the value if false. =IF(C2>=50,"Pass","Fail") shows Pass for 50 or more. Text results need double quotation marks; numbers, cell references and calculations do not, so =IF(J2>=50,0,$K$1) gives free delivery on larger orders and otherwise uses the charge in K1. Read operators carefully: > means greater than, so a value equal to the boundary gives the false result. A nested IF places a second IF in the false part of the first, giving three possible results, such as Distinction, Merit or Fail. Test the highest boundary first and close every bracket. VLOOKUP has four parts: the value to look for, the table range, the column number to return from, and FALSE for an exact match. If the value is not in the first column of the table, the result is #N/A. The table range is usually absolute, such as $G$20:$I$24, so that the formula can be replicated down a column without the table moving.

Checking input and setting up for others

Conditional formatting applies a format, such as a red fill or bold font, only to cells that meet a rule. Because the rule is applied automatically, the formatting changes when the data changes, which is more reliable than colouring cells by hand. Data validation controls input at the point of entry. A range check might allow only whole numbers from 1 to 16; a list check gives a drop-down of allowed values, which is quicker and avoids spelling mistakes. An input message can explain what to enter, and an error alert pops up if invalid data is typed. Validation can only check that data is sensible, not that it is correct. A template is a pre-designed workbook with headings, formatting, formulae and validation already in place, so other people only add their data. Protecting the formula cells stops users breaking them. Importing uses a wizard to bring in data from a database table, text file or web page; the file may need to be in CSV format first.

Modelling, charts and macros

A spreadsheet model represents a real situation with input values and formulae that act as rules. Changing an input, such as a price, makes every dependent formula recalculate, so the user can see the effect straight away. This is what-if analysis. In a sports camp model, raising the price from £35 to £38 but losing 15 bookings changes profit from £540 to £600, which helps the organiser decide. Models let managers test many options quickly, cheaply and without risk, but the predictions are only as reliable as the data and rules used. Charts make results easier to understand at a glance and to show in presentations. Choose the type to suit the data: column or bar to compare categories, line for trends over time, pie for parts of a whole. A good chart has a title, axis titles with units and a legend when there is more than one data series. A macro records a sequence of actions so it can be replayed with one click, saving time and making sure repeated tasks are done the same way every time.

T Key terms from the mark schemes

IF function
A function that tests a condition and returns one value if it is true and another if it is false, e.g. =IF(A1>60,"PASS","FAIL").
Credited: U1 2022 Q3, 2023 Q4, 2025 Q6

Nested IF
An IF function placed inside another IF function so that more than two outcomes are possible, e.g. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")).
Credited: U1 2021 Q3, 2022 Q3

VLOOKUP
Vertical lookup: a function that searches for a value in the first column of a table and returns a value from a specified column in the same row. It updates automatically if the source data changes.
Credited: U1 2019 Q1, 2022 Q3, 2025 Q6

Lookup table
The range of cells searched by VLOOKUP; the values to be matched are in its first column.

Conditional formatting
Formatting (e.g. fill or font colour) that is applied automatically to cells only when a specified condition is met, e.g. hours worked over 40 highlighted.
Credited: U1 2019 Q1, 2025 Q6

Data validation
An automatic check when data is entered that it is sensible and within acceptable limits, e.g. a range check of 1 to 16. It cannot check the data is correct.

Error alert
A message displayed by data validation when invalid data is entered, asking the user to re-enter valid data.

Template
A pre-designed spreadsheet with layout, formatting and formulae set up, used as the starting point for new documents with the same format.
Credited: U1 2021 Q3, 2023 Q4

Importing data
Bringing data into a spreadsheet from another application or file (e.g. a CSV file exported from a database) without retyping it.

Data modelling / what-if
Using a spreadsheet to represent a real situation and changing input values to predict outcomes and see the effect of the change, e.g. changing a price to see the effect on profit.
Credited: U1 2019 Q8

Charts and graphs
Visual representations of data (e.g. column, line, pie) used to compare values, show trends or show parts of a whole; they can be copied into other software.

Macro
A small program that stores a sequence of actions so they can be carried out automatically with one click or shortcut, saving time on repeated tasks.
Credited: U1 2021 Q3

Exam focus

IF questions appear almost every year (2021 Q3, 2022 Q3(f), 2023 Q4(d), 2025 Q6(e)). Marks are given for the function name, a correct condition, both results in the right order and correct punctuation, so write the whole formula, including the equals sign, commas and quotation marks around text. Nested IF has been asked for three results (2021 Q3, 2022 Q3), and 2025 combined IF with an absolute reference. VLOOKUP appears as a multiple-choice question (2019 Q1(h)) and as an explain question (2022 Q3(g), 2025 Q6(f)). Examiners reported weak answers, so learn to explain each part: what is looked up, where it is searched for, which column is returned and why FALSE is used. Chart questions (2019 Q8(d)–(e), 2023 Q4(e)–(f), 2024 Q4) ask for the cell ranges used and the best chart type with a reason. Templates (2021 Q3, 2023 Q4(g)), validation and macros (2021 Q3) and what-if modelling (2019 Q8(f)) need a clear description plus a benefit linked to the scenario.

Unit 1 2025 Q6	[10]	Spreadsheet features, ranges, SUM, IF with absolute reference, lookup functions
Unit 1 2024 Q5	[10]	Spreadsheet functions, CPU purpose and components, immediate-access store
Unit 1 2024 Q4	[13]	SUM, subtraction, wrap text, absolute references, chart ranges and type
Unit 1 2023 Q4	[11]	Merge and centre, SUM, formulae, absolute references, IF, charts, template
Unit 1 2022 Q3	[12]	Currency, merge and centre, formulae, ranges, nested IF, VLOOKUP

? Check yourself

1. Cell D2 contains a test mark. Write an IF function that displays 'Merit' if the mark is 70 or more, otherwise 'Pass'. [2]
2. Explain what the function =IF(A1>60,"PASS","FAIL") does. State the result if A1 contains 60. [3]
3. Cell B2 contains a student's mark. Write a nested IF function for C2 that displays "Distinction" for a mark of 80 or more, "Merit" for a mark of 65 or more and "Fail" otherwise. [3]

Answers: 1. =IF(D2>=70,"Merit","Pass") – correct condition D2>=70 (1) / both results in the right order with quotation marks (1) 2. Checks whether the value in A1 is greater than 60 (1) / if true displays PASS, otherwise displays FAIL (1) / A1 = 60 gives FAIL because 60 is not greater than 60 (1) 3. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")) – first condition and Distinction (1) / nested IF with B2>=65 and Merit (1) / Fail as final false value with correct brackets and quotes (1)

The CPU and the fetch–execute cycle

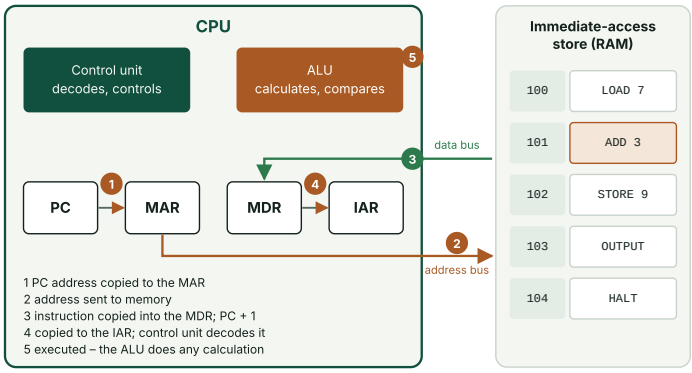
L13 FACT FILE

Specification: students should be able to

- Explain the purpose of the central processing unit (CPU)
- Describe the role of the following components of the CPU: the arithmetic logic unit (ALU), control unit and immediate-access store
- Describe the role the following play in the fetch-execute cycle: program counter, memory address register (MAR), memory data register (MDR), instruction address register (IAR) and ALU

CCEA fact file: Fact File 11: Computer Hardware, Computer Hardware 2

Key words: CPU, microprocessor, control unit, control signal, ALU, accumulator, immediate-access store, memory address, register, program counter, MAR, MDR, IAR, fetch, decode, execute, fetch–execute cycle



Inside the CPU: the fetch–execute cycle using the PC, MAR, MDR and IAR with main memory

What the CPU is for

The central processing unit (CPU) is the part of a computer that does the processing. It is a microprocessor, which means all of its circuits are built onto a single chip. Its purpose is to process data and carry out the instructions that make up computer programs. Every program, from a game on your phone to the till software in a shop in Belfast, is really a long list of simple instructions held in memory. The CPU works through these instructions one after another by repeatedly fetching an instruction from memory, decoding it to find out what it means, and then executing it. This repeating process is called the fetch–execute cycle. While doing this, the CPU also manages the rest of the computer by sending control signals to memory and to input and output devices. People often call the CPU the brain of the computer, but in an exam you must explain what it actually does: it processes data and executes program instructions.

Control unit, ALU and immediate-access store

The CPU has three main components. The control unit manages the running of instructions. It decides which instruction is next, fetches it, decodes it and then sends control signals to the other parts so that the instruction is carried out. It co-ordinates everything but does not do the calculations itself. The arithmetic logic unit (ALU) does the calculations and the logic. Arithmetic means adding, subtracting, multiplying and dividing. Logic means comparisons and the operators AND, OR and NOT, which let a program make decisions, such as checking whether a pupil has enough money on their canteen card. The ALU contains a special register called the accumulator, which holds the result of each calculation until it is stored back in memory. The immediate-access store (IAS) is the main memory. It holds the programs and data currently in use. Each memory location has a unique address, so the CPU can go straight to the instruction or data it needs.

Registers and what they hold

A register is a very small, very fast memory location inside the CPU, used for one particular job. Five matter at GCSE. The program counter (PC) holds the address of the next instruction to be fetched. Because most instructions run in order, it goes up by one after each fetch. The memory address register (MAR) holds the address of the memory location that is about to be read from or written to. The memory data register (MDR) holds the actual instruction or data that has just been fetched from memory, or that is about to be written to memory; all data going between the CPU and main memory passes through it. The instruction address register (IAR), as CCEA describes it, holds the instruction currently being decoded and executed; some textbooks call this the instruction register. Finally, the accumulator in the ALU holds the result of the latest calculation. A quick way to remember the difference: the MAR holds where, the MDR holds what.

The cycle step by step

In the fetch stage, the address held in the program counter is copied into the MAR. The control unit then reads that memory location, and the instruction stored there is copied into the MDR and then into the IAR. The program counter is increased by one so that it points to the next instruction. In the decode stage, the control unit works out what the instruction means, for example whether it is asking for a value to be loaded, two numbers to be added or a result to be stored. In the execute stage, the instruction is carried out. If it needs data from memory, the data's address goes into the MAR and the data comes back through the MDR. If it needs a calculation or comparison, the ALU does it and the result goes into the accumulator. The cycle then starts again with the next address in the program counter. A modern CPU repeats this cycle billions of times every second.

Key terms from the mark schemes

CPU
Central processing unit. A microprocessor whose purpose is to process data and carry out the instructions of programs, managing the functions of the computer.
Credited: Manages the functions of a computer by processing data and instructions

ALU
Arithmetic logic unit: the part of the CPU that carries out arithmetic calculations and logical operations and comparisons.
Credited: Arithmetic and logic unit

Control unit
The part of the CPU that fetches and decodes instructions and sends control signals to co-ordinate the other components.
Credited: Directs/co-ordinates the operations of the CPU

Immediate-access store
Main memory (RAM) holding the programs and data currently in use; each location has a unique address.
Credited: IAS; main memory

Register
A very small, very fast memory location inside the CPU used for a specific purpose.

Program counter
Register holding the address of the next instruction to be fetched; incremented after each fetch.
Credited: PC

MAR
Memory address register: holds the address of the memory location being read from or written to.

MDR
Memory data register: holds the data or instruction being transferred to or from memory.
Credited: Memory buffer register

IAR
Instruction address register (CCEA term): holds the instruction currently being decoded and executed.
Credited: Instruction register (IR)

Accumulator
Register in the ALU that holds the result of a calculation before it is stored in memory.
Credited: ACC

Fetch–execute cycle
The repeated process in which the CPU fetches an instruction from memory, decodes it and executes it.
Credited: Fetch–decode–execute cycle

Exam focus

This topic comes up nearly every year. You have been asked to explain the purpose of the CPU (2023 Q5, 2024 Q5), describe the ALU and control unit (2019 Q5(c)–(d)), explain registers and the cycle (2021 Q5, 2025 Q7(a)–(b)), and write a levelled answer on the ALU and immediate-access store (2022 Q4). Marks are given for each correct role, so 'describe' means say what the part does, not just name it. For the CPU's purpose, say it processes data and carries out program instructions; 'it is the brain' scores nothing on its own. Write every acronym in full at least once, as examiners regularly report lost marks for this. In fetch–execute questions, give the steps in the right order and name the register used at each step. Common mistakes are saying the ALU decodes instructions, mixing up the MAR and the MDR, and forgetting that the program counter is incremented. In levelled questions, use correct technical terms and link each point to the cycle.

Unit 1 2025 Q7	[8]	Fetch–execute cycle; input devices and uses
Unit 1 2024 Q5	[10]	Spreadsheet functions, CPU purpose and components, immediate-access store
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q4	[4]	ALU and immediate-access store (levelled)
Unit 1 2021 Q5	[6]	CPU, registers, speed, clock speed

? Check yourself

1. Explain the purpose of the central processing unit (CPU). [2]
2. State the three main components of the CPU. [3]
3. Describe the role of the arithmetic logic unit (ALU). Give an example of an operation it carries out. [3]

Answers: 1. It processes data (1) / it carries out (executes) the instructions of programs (1). Also accept: manages/controls the functions of the computer by sending control signals (1). 'Brain of the computer' alone = 0. 2. Control unit (1) / arithmetic logic unit (ALU) (1) / immediate-access store (IAS) or main memory (1). 3. Carries out arithmetic calculations (1) / carries out logical operations or comparisons, e.g. AND, OR, NOT, greater than (1) / valid example, e.g. adding two numbers or checking if a balance is below a limit (1). Accept: results held in the accumulator (1).

CPU performance and memory: RAM, ROM and cache

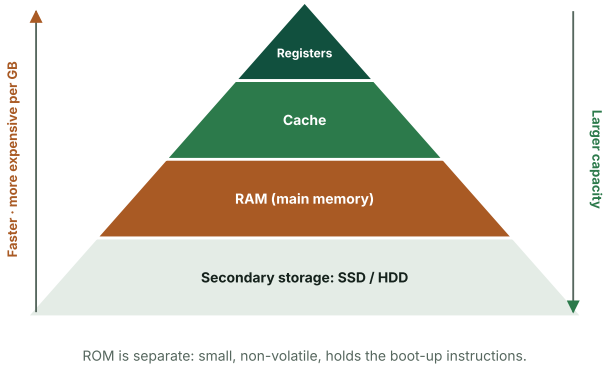
L14 FACT FILE

Specification: students should be able to

- Describe the impact of clock speed, cache size, and number of cores on CPU performance
- Explain the purpose of random access memory (RAM), read only memory (ROM) and cache
- Describe the role of ROM in the boot-up process
- Compare RAM, ROM and cache in terms of volatility, read/write access, size and speed

CCEA fact file: Fact File 11: Computer Hardware

Key words: clock speed, hertz, gigahertz (GHz), core, multi-core, parallel processing, cache, RAM, ROM, volatile, non-volatile, boot program, BIOS, virtual memory, performance



Memory hierarchy pyramid from registers to secondary storage

Three factors that affect CPU performance

The performance of a CPU means how quickly it can carry out instructions. The CCEA specification names three factors. The first is clock speed. The CPU contains a clock that sends out regular pulses, and the clock speed is the number of fetch–execute cycles the CPU can carry out each second. It is measured in hertz, and modern processors are measured in gigahertz: a 3 GHz processor can perform three billion cycles per second. The second factor is the number of cores. A core is a full processing unit: each one has a control unit, an ALU and a set of registers, so a quad-core processor can work on four instructions at the same time. The third factor is cache size. Cache is a small amount of very fast memory close to the CPU that holds frequently used instructions and data. When choosing a computer for a task, such as gaming or editing video, all three should be considered together, along with the amount of RAM.

Why the factors do not simply multiply

A higher clock speed means more instructions are processed each second, but it also produces more heat and uses more battery power. Also, if the other parts of the computer cannot supply data quickly enough, the CPU ends up waiting, so doubling the clock speed does not double the overall speed. In the same way, more cores only help if the software has been written to split its work into parts that can run in parallel. A simple program that does one thing at a time may use only one core, and the others sit idle. More cores are very useful when several programs are open, or for jobs that split up easily, such as rendering video. A larger cache means the CPU finds what it needs in the cache more often. Each time it does, it avoids a slower trip to RAM, so processing is faster. However, cache is expensive to make, which is why it is kept small.

RAM, ROM and cache

Internal memory is memory the CPU can access directly. Random access memory (RAM) is the main memory. Its purpose is to hold the operating system, programs and data that are currently in use, such as a web browser and the page you are reading. RAM can be read from and written to, and its contents change all the time. It is volatile, meaning everything in it is lost when the power is switched off, which is why files must be saved to storage. Read only memory (ROM) can be read but not normally changed. It is non-volatile, so it keeps its contents without power. Its purpose is to store the boot program, the instructions the computer needs as soon as it is switched on. Cache is the third type. Like RAM it can be read from and written to and it is volatile, but it is much smaller and much faster, and it sits inside or right beside the CPU.

Booting up and running out of RAM

When you press the power button, RAM is empty, so the computer cannot start from it. Instead, the CPU runs the start-up instructions stored in ROM, often called the BIOS. These instructions check that key hardware is working and then load the operating system from the hard disc or SSD into RAM. Once the operating system is in RAM, it takes over. While the computer is running, RAM can fill up if many programs are open. The operating system then uses part of the storage drive as extra memory, called virtual memory, moving data that is not being used right now out of RAM. Because storage drives are much slower than RAM, the computer slows down as data is swapped back and forth. This is why adding more RAM can make a computer faster: it reduces the need for virtual memory. The textbook says larger RAM makes the processor faster; more precisely, it stops the processor having to wait for slow storage.

T Key terms from the mark schemes

Clock speed
The number of fetch–execute cycles a CPU can carry out per second, measured in hertz (e.g. 3 GHz = 3 billion cycles per second).
`Credited: Processing cycles per second`

Cache
Small, very fast memory on or near the CPU that holds frequently used instructions and data, so they need not be fetched from slower RAM.
`Credited: Cache memory`

ROM
Read only memory: non-volatile memory whose contents cannot normally be changed, used to store the boot-up instructions (BIOS).

Non-volatile
Describes memory or storage that keeps its contents without power, such as ROM and SSDs.

Virtual memory
Part of a storage drive used as extra memory when RAM is full; it is much slower than RAM.

Core
An independent processing unit inside a CPU; each core has a control unit, an ALU and registers, so it can process instructions at the same time as other cores.
`Credited: Processor core`

RAM
Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use.
`Credited: Main memory`

Volatile
Describes memory that loses its contents when the power is switched off, such as RAM and cache.

Boot program
Instructions stored in ROM that run when the computer is switched on, checking hardware and loading the operating system into RAM.
`Credited: BIOS; bootstrap loader`

Multi-core processor
A CPU that contains two or more cores, so it can process several instructions in parallel.
`Credited: Dual-core / quad-core`

Exam focus

Performance and memory questions appear often. Clock speed and its units came up in 2021 Q5 and 2022 Q12(c). RAM and ROM have been tested in 2019 Q5(a) and 2024 Q6(e), and ROM's role in booting appeared in 2025 Q4. Cache came up in 2024 Q5(c) and as a matching term in 2025 Q13. For 'describe the impact' questions, give the cause and the effect: 'a higher clock speed means more fetch–execute cycles per second, so more instructions are processed per second'. For cache, the marks usually need two ideas: it is faster to access than RAM, and it holds frequently used data or instructions. For RAM versus ROM, give a matched pair of differences, such as 'RAM is volatile but ROM is non-volatile', rather than describing only one memory. Common mistakes are saying clock speed is measured in bytes, saying ROM holds the user's files, and saying that cache is part of the hard disc.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q4	[7]	Boot process, OS roles, storage conversion
Unit 1 2024 Q6	[13]	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes
Unit 1 2024 Q5	[10]	Spreadsheet functions, CPU purpose and components, immediate-access store
Unit 1 2022 Q12	[7]	WAN, encryption, firewall, processor performance

? Check yourself

1. State the unit used to measure clock speed. [1]
2. Describe the impact of increasing the clock speed on the performance of a CPU. [2]
3. A computer in the school library has a 3.2 GHz processor. State how many cycles per second this is, written in full. [1]

Answers: 1. Hertz / Hz / gigahertz (GHz) (1). 2. Higher clock speed means more fetch–execute cycles per second (1) / so more instructions are processed per second and performance is faster (1). 3. 3,200,000,000 (three thousand two hundred million / 3.2 billion) cycles per second (1).

Input, output and storage devices

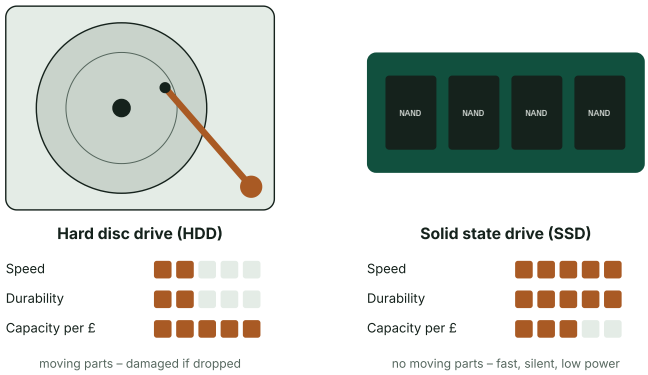
L15 FACT FILE

Specification: students should be able to

- Describe the characteristics, typical uses, and advantages and disadvantages of: microphone, mouse, graphics digitiser, touch screens, speakers, printers (laser and 3D), hard disc drive (HDD), high definition (HD) storage media and solid state drive (SSD)
- Describe other solid-state storage: USB flash drives, memory cards and smart cards
- Compare storage devices using capacity, access speed, portability, durability and cost
- Recommend and justify suitable devices for a given scenario

CCEA fact file: Fact File 11: Computer Hardware

Key words: input device, output device, storage device, microphone, voice recognition, graphics digitiser, stylus, touch screen, laser printer, toner, 3D printer, CAD, HDD, SSD, flash memory, Blu-ray, capacity, durability



Hard disc drive with spinning platter compared with a solid state drive made of memory chips

Devices and why they are chosen

Hardware devices are grouped by what they do. Input devices get data into a computer: a microphone captures sound, a mouse controls the pointer, a graphics digitiser captures freehand drawing with a stylus, and a touch screen lets users select options by touching the display. Output devices present information: speakers produce sound, a laser printer produces high-quality pages, and a 3D printer builds solid objects. Storage devices keep data permanently, even when the power is off: hard disc drives store data magnetically, solid state drives use flash memory, and high-definition optical discs such as Blu-ray use a laser. A touch screen is unusual because it is both an input and an output device. No device is best in every situation. A device is chosen by matching its features to the task, the users and the environment. A microphone is ideal for a gamer chatting to teammates, but voice input would be a poor choice in a noisy school canteen. Exam questions nearly always give a scenario, so every advantage you give should be linked to it.

Input devices

A microphone turns sound into an electrical signal that is converted into digital data. It is used for voice recognition, such as dictating text or giving commands to a smart speaker, as well as for video calls, recording music and online gaming headsets. Speaking is faster than typing and is hands-free, but background noise reduces accuracy and sound files need much more storage than text. A mouse uses sensors to detect movement and moves the on-screen pointer. It is easy to learn, but needs a flat surface, and experienced users often find keyboard shortcuts quicker. A graphics digitiser is a flat pad used with a stylus, much like a pencil on paper. It gives more natural and more accurate drawing than a mouse, which suits artists and designers, and it can also capture signatures. However, it is more expensive than a mouse and poor for selecting menus. A touch screen has a touch-sensitive glass surface. A touch changes the electrical signal, and the position is worked out as x, y co-ordinates. Touch screens need little skill and are quick, but they get dirty and can only show a limited number of options.

Output devices

Speakers turn electrical signals from the sound card into sound. They are used for music, video, games and alert sounds, and they help visually impaired users by reading text aloud. Their main drawback is that they can disturb others nearby, so headphones may be better in shared spaces. A laser printer receives the page as data. A laser beam draws the page onto a drum as a pattern of static charge, powdered toner sticks to the charged areas, and the toner is transferred to paper and fixed with heat. Laser printers are fast for large volumes and give sharp text and graphics, which is why schools and offices use them. However, colour models are expensive to buy and run, and because they are non-impact printers they cannot print multipart carbon-copy forms. A 3D printer builds a solid object one layer at a time from a CAD design, using materials such as plastic, metal or ceramics. It is ideal for prototypes and custom items such as prosthetic hands, but it is slow for large objects and not yet economical for mass production.

Storage devices

A hard disc drive (HDD) contains rigid magnetic discs, called platters, sealed in a case to keep dust out. Each surface is split into circular tracks and the tracks into sectors, and read/write heads move across the spinning discs to reach data. HDDs offer very large capacities at the lowest cost per gigabyte, but their moving parts can fail or be damaged if dropped, and access is slower than flash storage. A solid state drive (SSD) uses flash memory with no moving parts. It is non-volatile, much faster, quiet, light, uses less power and survives knocks, but costs more per gigabyte. Other flash storage includes USB drives, memory cards for phones and cameras, and the smart-card chips in bank cards and door entry cards. High-definition storage media, such as Blu-ray, are optical discs. A blue-violet laser has a shorter wavelength than the red laser in a DVD drive, so the pits are smaller and packed more tightly, and a dual-layer disc holds 50 GB. They suit HD films and games, but scratch easily.

T Key terms from the mark schemes

Input device
Hardware used to enter data into a computer, such as a microphone, mouse or graphics digitiser.

Output device
Hardware used to present information from a computer, such as speakers or a printer.

Graphics digitiser
An input device made of a flat pad and a stylus, used to capture freehand drawings or signatures.
`Credited: Graphics tablet`

Touch screen
A display with a touch-sensitive surface that works as both an input and an output device.
`Credited: Touchscreen`

Laser printer
An output device that uses a laser to form the page on a charged drum, which attracts toner that is fused onto paper; fast and high quality.

3D printer
An output device that builds a solid object layer by layer from a computer-aided design (CAD) model.
`Credited: Additive manufacturing`

Hard disc drive (HDD)
Magnetic storage using spinning platters and moving read/write heads; large capacity at low cost per gigabyte.
`Credited: HDD`

Solid state drive (SSD)
Non-volatile flash storage with no moving parts: faster, more durable and lower power than an HDD, but more expensive per gigabyte.
`Credited: SSD`

Optical storage
Storage read and written by a laser, such as Blu-ray, the HD storage media named in the specification.
`Credited: HD storage media`

Flash memory
Non-volatile electronic memory with no moving parts, used in SSDs, USB drives and memory cards.
`Credited: Solid-state storage`

Smart card
A plastic card with an embedded chip that stores a small amount of data, such as a chip-and-PIN bank card.

Exam focus

Devices are tested often and usually in a context. Input devices and their uses appeared in 2023 Q5(e), 2024 Q9(a) and 2025 Q7(c). Laser printer and SSD advantages were tested in 2021 Q7, and an SSD advantage in 2019 Q5(a). 3D printing and graphics digitisers came up in 2022 Q5, and optical storage and capacity terms in 2022 Q2 and the 2022 Q1 multiple choice. Marks are given for a valid point that is clearly explained and linked to the scenario. 'An SSD is faster' may get one mark; 'an SSD has no moving parts, so the inspector's laptop is less likely to be damaged if dropped' gets more. Make your points comparative, such as 'lower cost per gigabyte than an SSD'. Do not give the same idea twice in different words, and avoid vague answers such as 'it is better' or 'it is easier'. Learn the key facts: 3D printers work layer by layer, laser printers use toner, HDDs are magnetic and SSDs have no moving parts.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q7	[8]	Fetch–execute cycle; input devices and uses
Unit 1 2024 Q9	[7]	Input devices, health risks, gaming and internet risks
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q5	[6]	3D printing, utilities and security, graphics digitiser

? Check yourself

1. State whether each of the following is an input device, an output device or both: microphone, speakers, touch screen. [3]
2. Describe one advantage and one disadvantage of using a touch screen to take orders in a fast-food restaurant. [2]
3. Describe what a graphics digitiser is and give a typical use. [2]

Answers: 1. Microphone – input (1) / speakers – output (1) / touch screen – both (1). 2. Advantage, one from: customers or staff need little training / quick to select items / no separate keyboard or mouse (1). Disadvantage, one from: screen gets dirty or greasy, a hygiene issue (1) / limited number of options on screen at once / can be damaged with heavy use. 3. A flat pad/tablet used with a stylus to capture freehand drawing (1). Use: graphic design / artwork / animation / capturing signatures (1).

Networks: LAN, WAN, internet, hardware and topologies

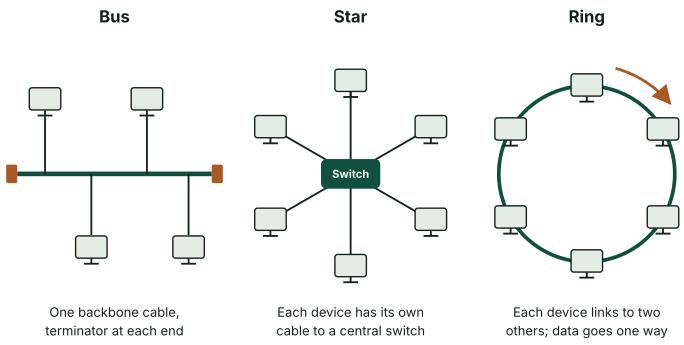
L16 FACT FILE

Specification: students should be able to

- Describe the main features of a local area network (LAN) and a wide area network (WAN)
- Describe the advantages and disadvantages of using a network in an organisation
- Describe the difference between the internet, the World Wide Web, an intranet and the Internet of Things
- Describe the function of a network interface card, network cables, a switch and a router
- Describe the bus, star and ring network topologies, including their advantages and disadvantages

CCEA fact file: Network Technologies 1, Network Technologies 2

Key words: network, standalone computer, LAN, WAN, file server, internet, World Wide Web, URL, intranet, Internet of Things, NIC, switch, router, IP address, topology, bus, terminator, star, ring



Bus, star and ring network topologies

Networks: the big picture

A network is a group of computers and devices linked by cable or wirelessly so they can share resources and communicate. A computer that is not linked to anything is called a standalone computer. A local area network (LAN) covers a small area, such as one school or office, and its hardware is usually owned by the organisation. A wide area network (WAN) covers a large geographical area and relies on telecommunications links such as fibre optic cables or satellites, often rented from other companies. The internet is the biggest WAN of all. Organisations use networks because users can log on anywhere, share printers and files, and have their data backed up and protected centrally. Networks also have costs: hardware must be bought, a technician is needed, and viruses can spread from one computer to all the others.

Internet, web, intranet and IoT

The internet and the World Wide Web are not the same thing. The internet is the global network of networks: the cables, routers and connections that carry data. The World Wide Web is one service that runs on the internet. It is made up of web pages, written in HTML and joined by hyperlinks, which are stored on web servers. Each site has a unique URL and pages are sent using HTTP or HTTPS. An intranet uses the same technology as the internet but is private. Only authorised users who log on with a username and password can use it, so it is useful for sharing staff documents, forms and messages. The Internet of Things (IoT) describes everyday objects, such as fitness watches, smart thermostats and video doorbells, that contain sensors and connect to the internet to send and receive data.

Network hardware

Every computer on a network needs a network interface card (NIC). This circuit board lets the computer connect to the network and gives it a port for a cable. Laptops and phones have a wireless NIC instead. Network cables physically link each NIC to another device. Most are copper twisted-pair cables in categories such as CAT5e and CAT6, where CAT6 can carry data faster. A switch is the central connection point in a LAN. It checks where each piece of data is going and forwards it only to the intended computer, using MAC addresses. A router connects separate networks, such as a school LAN and the internet. It reads the IP address on data and sends it along the best route to its destination. Many routers also include a firewall to filter traffic.

Topologies compared

A topology is the layout of a LAN. In a bus network every computer joins one backbone cable, with a terminator at each end to stop signals reflecting back. It is cheap and needs little cable, but if the backbone breaks the whole network stops, and collisions slow it down when it is busy. In a star network each computer has its own cable to a central switch (or server). A broken cable only affects one computer and it is easy to add more, but more cable is needed and the whole network fails if the central device fails. In a ring network each computer links to two neighbours to form a loop and data moves in one direction. There are no collisions, but one broken link stops the ring, and adding a computer means breaking the loop.

T Key terms from the mark schemes

LAN
Local area network: a network covering a small geographical area such as one building or site, usually owned by the organisation.
`Credited: Network in a small area, e.g. a school`

WAN
Wide area network: a network covering a large geographical area that uses telecommunications links such as fibre optic cables or satellites.
`Credited: Network over a large geographical area`

Standalone computer
A computer that is not connected to a network, so its files and devices cannot be shared.

Internet
The global network of networks: the hardware and connections that carry data between computers worldwide.
`Credited: A worldwide WAN`

World Wide Web
A collection of linked web pages stored on web servers and accessed over the internet using a browser.
`Credited: WWW`

Intranet
A private network owned by an organisation that uses internet technology but can only be accessed by authorised users who log on.
`Credited: Private network for an organisation's members`

Internet of Things (IoT)
Everyday physical devices with sensors and connectivity that send and receive data over the internet, e.g. fitness watches and smart thermostats.
`Credited: Everyday devices connected to the internet`

NIC
Network interface card: a circuit board in a computer that lets it connect to a network and provides a port for the network cable.
`Credited: Network adapter`

Switch
A device that connects computers on a LAN and forwards data only to the intended computer.
`Credited: Central connection point that sends data to the correct device`

Router
A device that connects networks, such as a LAN and the internet, and forwards data along the best route using IP addresses.
`Credited: Connects networks and directs packets`

Bus topology
A layout where all devices are attached to a single backbone cable with a terminator at each end.

Terminator
A device at each end of a bus backbone that absorbs the signal so it does not bounce back along the cable.

Star topology
A layout where each device is connected by its own cable to a central switch or server.

Ring topology
A layout where each device connects to two others in a closed loop and data travels in one direction.

Exam focus

This lesson appears on almost every Unit 1 paper. Topologies are the favourite: bus, star and ring diagrams or descriptions came up in 2019 Q6(b)–(d), 2022 Q7(a), 2023 Q7(c) and 2024 Q6(b). For a bus network, name the backbone and say what a terminator does; for a disadvantage, say the whole network fails if the backbone fails. Switch and router were asked in 2021 Q6 and 2023 Q7(b). Keep them separate: a switch forwards data to the right computer within a LAN, a router connects networks and uses IP addresses to route data. In 2025 Q8(a) many candidates could not expand IoT, so learn 'Internet of Things' and have an example ready. Internet versus WWW (2022 Q8(c), 2025 Q8(b)) needs both halves: network of networks versus pages accessed over it. 'Distinguish' questions need a comparison, not two separate definitions.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q8	[10]	IoT, Internet vs WWW, uses of networks, networking concepts
Unit 1 2024 Q6	[13]	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes
Unit 1 2023 Q7	[8]	LAN, router and switch, bus diagram, NIC
Unit 1 2023 Q6	[5]	Communication technologies; IoT

? Check yourself

1. Southern Regional College links the computers in each of its campuses, and also links its Newry, Armagh and Banbridge campuses together. Describe two differences between the network inside one campus (a LAN) and the network joining the campuses (a WAN). [4]
2. Explain the difference between the internet and the World Wide Web. [2]
3. A grammar school in Coleraine has an intranet for its pupils and staff. Explain how the school's intranet differs from the internet. [4]

Answers: 1. Any two differences, each fully described (2 marks each): LAN covers a small geographical area such as one building/site (1) whereas a WAN covers a large geographical area such as countries/the world (1) / LAN hardware is usually owned by the organisation (1) whereas a WAN uses telecommunications links rented from other companies (1) / LAN uses network cables or Wi-Fi (1) whereas a WAN uses links such as fibre optic cables, satellites or telephone lines (1) 2. The internet is a global network of networks / the hardware and connections that carry data (1) / the WWW is a collection of web pages/websites that are accessed over the internet using a browser (1) 3. Internet: public / accessible to anyone with a communications link (1) / global network of networks (1) / Intranet: private network owned by an organisation (1) / only staff and pupils who log on with their own username and password can use it (1) / uses the same technology/protocols as the internet (1) / used to share internal information such as school resources and notices (1) – max 2 for each side, comparison needed for full marks

Network communication technologies

L17
FACT FILE

Specification: students should be able to

- Describe Wi-Fi, Bluetooth, optical fibre and mobile communication technology (4G and 5G)
- Evaluate the effectiveness of each communication technology in terms of speed, range, reliability, security and cost
- Compare wireless communication with optical fibre communication
- Select and justify a suitable communication technology for a given scenario

CCEA fact file: Network Technologies 2

Key words: Wi-Fi, wireless router, radio waves, Mbps, Bluetooth, pairing, optical fibre, bandwidth, repeater, interference, 4G, 5G, latency, coverage, encryption



Typical range of Bluetooth, Wi-Fi, 4G, 5G and optical fibre

Overview of communication technologies

Networks can send data along cables or through the air. Wi-Fi uses radio waves and a wireless router so devices can join a network without cables. Bluetooth also uses radio signals, but it links two nearby devices directly, over about 10 metres, after they have been paired. Optical fibre sends data as pulses of light along thin glass strands, giving very high bandwidth over long distances. Mobile technologies, 4G and 5G, use radio masts to give wireless internet across a wide area, so people can stay connected while they move. To evaluate these technologies, think about five things: speed (bandwidth), range, reliability, security and cost. No single technology is best for everything, so the right choice depends on the situation.

Wi-Fi and Bluetooth

Wi-Fi lets laptops, phones and smart devices connect to a network through a wireless router or access point. It is cheap and quick to set up because no cables are needed, and users can move around a building. However, the signal gets weaker the further you are from the router, and walls and other devices cause interference. Speed also falls when many people share one access point. Older Wi-Fi standards reach about 54 Mbps, while modern standards are much faster. A Wi-Fi network should have a strong password and WPA2 or WPA3 encryption. Bluetooth is designed for short distances. It uses very little power and needs no router, which makes it ideal for earbuds, smartwatches, wireless mice and hands-free car kits. Its range is short, it is slower than Wi-Fi and only a few devices can connect at once.

Optical fibre

An optical fibre cable is a bundle of very thin glass strands. Data travels along each strand as pulses of light, and because each strand carries its own signal, a huge amount of data can be sent at the same time. This gives very high bandwidth and speed. Light is not affected by electrical interference, so fibre is reliable near machinery, and it is very hard for a hacker to tap into the cable without being noticed, so data is secure. Signals can travel many kilometres with little loss, although repeaters are needed to boost them over very long distances. The main drawback is cost: installing fibre often means digging up roads and paying specialists. The glass can also break if the cable is bent too sharply. Fibre is used for internet backbones, undersea cables and full-fibre broadband to homes.

4G and 5G mobile technology

Mobile phone networks divide an area into cells, each served by a mast. Phones send and receive data from the nearest mast using radio waves. 4G gives high-speed data, good enough for streaming HD video and making video calls, and it lowered the cost of mobile data. Because masts are expensive, coverage is best in towns and weaker in rural areas. Using mobile data also drains the battery faster than Wi-Fi. 5G is faster again and has much lower latency, which means less delay. It can support many more devices in one place, which helps the Internet of Things, cloud gaming and connected vehicles. The highest-speed 5G signals do not travel as far and are blocked by buildings, so many more masts are needed and coverage is still growing.

Key terms from the mark schemes

Wi-Fi
Wireless technology that uses radio waves and a wireless router or access point to connect devices to a network without cables.
Credited: Wireless fidelity; wireless LAN access

Bluetooth
Short-range, low-power wireless technology that connects two nearby devices directly after they have been paired.
Credited: Short-range wireless link between devices

Optical fibre
Cable made of thin glass strands that carries data as pulses of light, giving very high bandwidth and immunity to electrical interference.
Credited: Fibre optic

4G
Fourth-generation mobile technology giving high-speed wireless data through a network of masts.

5G
Fifth-generation mobile technology with faster speeds, much lower latency and capacity for more devices than 4G, but needing more masts.
Credited: Fast, low-latency mobile network

Bandwidth
The amount of data that can be transmitted in a given time, measured in bits per second.
Credited: Data transfer rate

Latency
The delay between data being sent and being received.
Credited: Lag

Pairing
Linking two Bluetooth devices so they recognise each other and can exchange data.
Credited: Synchronising

Repeater
A device that boosts a signal so it can travel over longer distances.

Interference
Unwanted signals or obstacles that weaken or corrupt transmitted data.

Exam focus

Questions on this lesson often ask you to match technologies to uses or descriptions, as in 2021 Q11, 2022 Q8(b) and 2023 Q6(a). Learn one clear feature and one typical use for each technology: Bluetooth for short-range links such as earbuds, Wi-Fi for wireless access to a network in a building, fibre for very fast secure links, 5G for fast, low-latency mobile data. In 2022 Q8(d) candidates judged true/false statements about fibre optic, so know the facts exactly: light pulses, glass strands, high bandwidth, not affected by electrical interference, hard to tap, expensive to install. When asked to evaluate or compare, give both an advantage and a disadvantage and use comparative words such as 'faster than' or 'more secure than'. Vague answers like 'it is better' or 'it is quick' earn nothing unless you say what it is compared with.

Unit 1 2023 Q6	[5]	Communication technologies; IoT
Unit 1 2022 Q8	[11]	IoT, Wi-Fi and Bluetooth, Internet vs WWW, fibre optic
Unit 1 2021 Q11	[8]	Key term matching

Check yourself

1. Describe Bluetooth technology and give one typical use. [3]
2. A family in Larne connect their phones, tablets and smart TV to the internet using Wi-Fi. State two features of the wireless technology they are using. [2]
3. Give two advantages and one disadvantage of optical fibre as a communications medium. [3]

Answers: 1. Short-range wireless technology (1) / uses radio signals to connect two devices directly (1) / devices must be paired/synchronised (1) / low power (1) / no router needed (1) – max 2 / use, e.g. wireless headphones, hands-free car kit, wireless mouse, smartwatch to phone (1) 2. Any two: uses radio waves (1) / needs a wireless router or access point (1) / no network cables required (1) / devices have a built-in wireless adapter (1) / signal strength depends on distance from the router (1) / should be protected by a password/encryption (1) / speed depends on the Wi-Fi standard (1) 3. Advantages (any two): very high bandwidth/speed (1) / not affected by electrical interference (1) / very hard to tap so data is secure (1) / carries data long distances with little loss (1) / Disadvantage (any one): expensive to install (1) / fragile – can break if bent (1) / needs repeaters over very long distances (1) / specialist skills needed to repair (1)

Cybercrime and malware

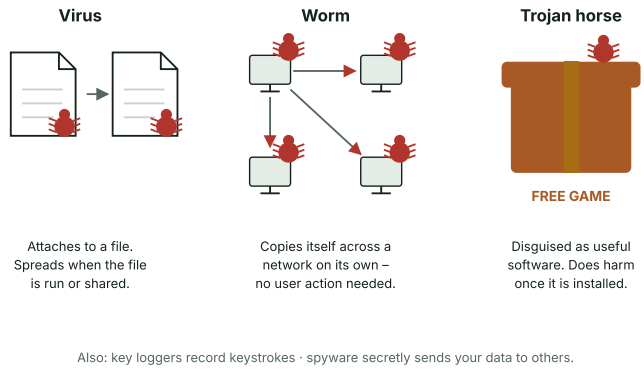
L18
FACT FILE

Specification: students should be able to

- Define the term cybercrime and give examples of threats to cybersecurity, including hacking, pornography, cyber stalking, data theft, denial of service, digital forgery, cyber defamation, spamming and phishing
- Define the term malware and describe the following forms of malware: virus, Trojan horse, worm, key logger and spyware
- Describe how malware can get onto a computer system and how anti-virus software helps to prevent it

CCEA fact file: Cyberspace

Key words: cybercrime, cybersecurity, hacking, cyber stalking, data theft, identity theft, denial of service, digital forgery, cyber defamation, spam, phishing, malware, virus, Trojan horse, worm, key logger, spyware, anti-virus software



How a virus, a worm and a Trojan horse spread

Cybercrime and cybersecurity

Cybercrime is any crime that is carried out using a computer, a network or the internet. Sometimes the computer is the target of the crime, for example when a hacker breaks into a council database or when a website is knocked offline. Sometimes the computer is simply the tool, for example when it is used to stalk someone or to forge a document. The people responsible are called cyber criminals, and they may be after money, personal data, revenge or simply disruption. Cybersecurity means the methods used to protect computers, networks and data against these threats. The CCEA specification lists nine threats you must be able to describe: hacking, pornography, cyber stalking, data theft, denial of service, digital forgery, cyber defamation, spamming and phishing. It also lists five forms of malware: virus, Trojan horse, worm, key logger and spyware. For each one you should be able to say what it is, give an example and explain the harm it causes.

Threats aimed at organisations

Hacking is gaining unauthorised access to a computer system, and it is an offence under the Computer Misuse Act 1990. A hacker might read, steal, change or delete data. Data theft is the stealing of records such as customer names, addresses and card numbers. The stolen details are often used for identity theft, where a criminal pretends to be the victim to apply for credit cards or loans. A denial of service attack floods a web server with so many requests that genuine users cannot get in, so an online shop loses sales and customers lose trust. Spamming is the sending of bulk junk messages; it wastes time, fills inboxes and uses bandwidth. Phishing uses emails, texts or fake websites that look as if they come from a trusted organisation, such as a bank or delivery company. They try to trick the person into clicking a link and typing in personal details, which are then used for fraud.

Threats aimed at people

Some cybercrimes harm individuals directly. Cyber stalking is using technology to repeatedly harass, follow or frighten someone, for example by sending threatening messages or tracking their location from social media posts. Cyber defamation is posting false statements online that damage the reputation of a person or a business, such as untrue reviews claiming a restaurant is unhygienic. Digital forgery is deliberately altering digital content, like photos, certificates or passport scans, to deceive people; deepfake videos are a modern example. Pornography is a threat in two ways. Sharing illegal or indecent images is a crime, particularly images of children or images shared without consent. In a workplace, adult websites are also a common hiding place for malware, so a member of staff who visits them can put the organisation's network and data at risk. This is one reason schools and companies filter which websites can be visited.

Malware and how it spreads

Malware is short for malicious software: software written to damage or disrupt a computer, or to gain unauthorised access to it or the data on it. It usually reaches a computer without the user meaning to install it, through email attachments, links in messages, downloads from untrusted websites, fake apps or infected USB memory sticks. A virus attaches itself to a file or program and runs when that file is opened, sometimes waiting for a trigger such as a date; it can corrupt or delete data and copies itself into other files. A worm does not need a host file. It copies itself and travels across a network by itself, for example by emailing itself to every contact, and it eats up bandwidth. A Trojan horse pretends to be useful software so the user installs it; it might then delete files or open a back door for a hacker. Anti-virus software, updated regularly, scans for and removes known malware.

Key loggers and spyware

Key loggers and spyware are both designed to gather information without the user knowing. A key logger records every key the user presses and saves it in a log file, which is later sent to the criminal. By reading the log, the criminal can find usernames, passwords and card numbers typed into websites and then log in to the victim's accounts or spend their money. Spyware is installed secretly, often bundled with free software downloaded from the internet. It monitors wider activity, such as the websites visited and items bought, and sends this information to a third party. The data may be sold to advertisers or used for fraud. Spyware also uses memory and processing power, so the computer can become slow or programs may crash. Some employers legally use monitoring software on work computers, but they must tell staff, which is the key difference from spyware.

T Key terms from the mark schemes

Cybercrime
Any crime committed using a computer, a network or the internet.
`Credited: Using a computer to commit a crime`

Hacking
Gaining unauthorised access to a computer system or its data, for example to steal, change or delete data.
`Credited: Unauthorised access`

Phishing
Emails, texts or websites that pretend to come from a trusted organisation in order to trick people into giving personal details such as passwords or bank details.

Denial of service
An attack that floods a server or website with requests so that it cannot respond to genuine users.
`Credited: DoS`

Cyber stalking
Using technology to repeatedly harass, track or frighten a person.

Cyber defamation
Posting untrue statements online that harm the good name of a person or an organisation.

Digital forgery
Deliberately and falsely altering digital content, such as photos or documents, to deceive.

Spamming
Sending bulk unwanted messages, usually adverts or scams, to many people.
`Credited: Junk email`

Identity fraud / theft
Using another person's personal details to pretend to be them, for example to obtain credit or goods.

Malware
Malicious software designed to damage or disrupt a computer system or to gain unauthorised access to it or its data.

Virus
Malware that attaches itself to a file or program and spreads when the file is run, copying itself and damaging or deleting data.

Worm
Self-replicating malware that spreads across a network by itself without needing a host file, using up bandwidth.

Trojan horse
Malware disguised as useful or legitimate software so the user installs it; it may delete files or open a back door for hackers.

Key logger
Malware that secretly records the keys a user presses and sends the log to a criminal, revealing passwords and card details.
Credited: Keystroke logger

Spyware
Software installed secretly that monitors a user's activity and sends the information to a third party.

Anti-virus software
Software that scans for, quarantines and removes known malware; it must be updated regularly to recognise new threats.
Credited: Virus checker

Exam focus

Cybercrime and malware come up every year. Threats have been tested in 2021 Q10(b) and 2025 Q11, and 2024 Q8 was a six-mark levelled question on phishing and denial of service. Malware was tested in 2021 Q9 (Trojan horse), 2022 Q7, 2023 Q8 and 2024 Q6, which asked how malware gets onto a computer, and many key-term matching questions (2021 Q11, 2024 Q10, 2025 Q13) include malware types. For 'describe' questions, give what the threat is plus what it does or its effect: 'phishing is a fake email' earns little, but adding that it pretends to be from a trusted source to trick the user into giving personal details earns full marks. Examiners have said that 'malicious software' on its own is not enough. For difference questions, compare both items directly, for example a worm spreads by itself whereas a virus needs a host file. In levelled questions, describe each threat, explain its impact and use the scenario.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q11	[6]	Cybercrime types; malware
Unit 1 2024 Q10	[8]	Key term matching
Unit 1 2024 Q8	[6]	Two cyber threats: phishing and denial of service (levelled)
Unit 1 2024 Q6	[13]	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes

? Check yourself

1. Define the term cybercrime. [1]
2. Describe phishing. [2]
3. A Newry sports shop sells kit through its website. Describe how a denial of service attack would affect the shop. [2]

Answers: 1. A crime committed using a computer, a network or the internet (1). 2. An email/text/website that pretends to be from a trusted/genuine organisation, e.g. a bank (1) / that tricks the user into giving personal details such as passwords or card numbers, e.g. by clicking a link (1). 3. The website/server is flooded with requests so it cannot respond / goes offline (1) / so genuine customers cannot access the site, so the shop loses sales/reputation (1).

Protecting networks and data; protocols

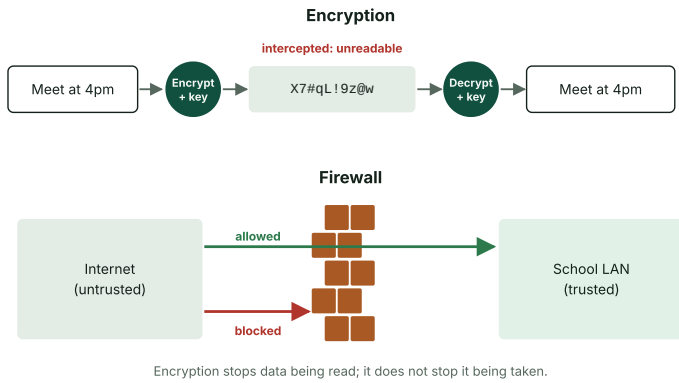
L19 FACT FILE

Specification: students should be able to

- Explain how networks and data can be protected using encryption, passwords, levels of access, backup and firewalls
- Describe the role of a protocol in data transfer
- Describe the purpose of the following protocols: File Transfer Protocol (FTP), HyperText Transfer Protocol (HTTP) and HyperText Transfer Protocol Secure (HTTPS)

CCEA fact file: Network Security and Data Transfer

Key words: encryption, decryption, encryption key, user ID, strong password, two-factor authentication, levels of access, read only, read and write, backup, restore, firewall, protocol, TCP/IP, FTP, HTTP, HTTPS, status code



Encryption makes intercepted data unreadable; a firewall filters network traffic

Protecting a network

Every organisation that stores data, from a primary school to a hospital trust, must stop that data being lost, stolen, read or changed by the wrong people. Under UK GDPR and the Data Protection Act 2018, organisations must keep personal data secure. The CCEA specification names five ways to protect networks and data: encryption, passwords, levels of access, backup and firewalls. Each one protects against a different problem. Passwords and levels of access control who can get in and what they can do once inside. Firewalls guard the boundary between the network and the internet. Encryption makes data useless to anyone who intercepts or steals it. Backups make sure data can be recovered if it is lost or corrupted. No single method is enough on its own, so organisations use them together. Data travelling across networks also follows protocols, the agreed rules that let different devices communicate, and some protocols, such as HTTPS, include encryption.

Encryption

Data sent across a network can be intercepted, and laptops and memory sticks can be lost or stolen. Encryption protects against this by using software, an algorithm and a key to scramble the data into cipher text before it is stored or sent. Anyone who intercepts the cipher text cannot understand it. Only someone with the correct key can decrypt it, turning it back into readable plain text. A simple example is a Caesar cipher, where each letter is shifted a set number of places along the alphabet, but real encryption uses very long keys and complex mathematics so it cannot be cracked by guessing. Encryption is used to secure Wi-Fi networks, HTTPS websites, messaging apps and staff laptops. A common exam mistake is to say encryption stops data being intercepted or stolen. It does not; it means that if the data is intercepted or stolen, it cannot be read or understood.

Passwords and levels of access

Each network user is given a unique user ID and a password. When they log on, both are checked against a secure database of users; if they do not match, access is refused. Passwords are shown as asterisks so others cannot read them, accounts can be locked after several failed attempts, and users may be made to change passwords regularly. A strong password is long, mixes upper and lower case letters, digits and symbols, and is not a dictionary word or personal detail such as a pet's name. Two-factor authentication adds a second check, for example a code sent to a phone. Levels of access then control what each user can do once logged on: no access, read only, read and copy, or read and write. Rights are stored against each user ID and checked whenever a file is opened. This keeps confidential data private, prevents accidental or deliberate changes, and limits the damage if one account is hacked.

Backups and firewalls

A backup is a copy of data stored separately from the original so that it can be restored if the original is lost, corrupted, deleted or encrypted by ransomware. Network backups are usually scheduled to run automatically, often overnight, and are kept in a safe place away from the server, for example on an external drive kept off-site or in cloud storage, so that a fire or theft does not destroy both copies. When needed, restore software copies the data back. A firewall is hardware or software that sits between a network and the internet. It monitors and filters all data entering or leaving the network and checks it against a set of rules called the security policy. Data that meets the rules is allowed through; data that does not is blocked. This keeps hackers out, can stop some viruses and spam, and can stop users visiting unsuitable websites. Firewalls also log blocked attempts so the network manager can investigate.

Protocols: FTP, HTTP and HTTPS

A protocol is an agreed set of rules for sending and receiving data. Sender and receiver must use the same protocol, rather like two people needing a shared language. Protocols set out the format of the data, how it is addressed, how errors are checked and how communication begins and ends, so devices from different manufacturers can work together. The internet uses TCP/IP. FTP, the File Transfer Protocol, is used to upload and download files between computers and a file server, for example when a web designer uploads a club's website files; users normally log in with a username and password. HTTP, the HyperText Transfer Protocol, is used by browsers to request web pages from web servers; if a page is missing the server returns a status code such as 404. HTTPS is HTTP with encryption, so intercepted data such as card numbers is meaningless. Browsers show a padlock for HTTPS sites.

T Key terms from the mark schemes

Encryption
Scrambling data using an algorithm and a key so that it cannot be understood by anyone who does not have the key.
`Credited: Encoding data so it is unreadable if intercepted`

Decryption
Turning encrypted data back into readable form using the correct key.

Firewall
Hardware or software that monitors and filters data entering or leaving a network, blocking traffic that does not meet security rules.

Access rights / levels of access
Permissions linked to a user ID that control what each user can see and do, e.g. no access, read only, read and write.
`Credited: Access levels`

Read only
An access level where a user can view data but cannot change it.

Strong password
A long password that mixes upper and lower case letters, digits and symbols and is not a dictionary word or personal detail.

Two-factor authentication
A logon that needs two different forms of verification, such as a password and a code sent to a phone.
`Credited: 2FA`

Backup
A copy of data stored separately from the original so it can be restored if the original is lost or corrupted.

Protocol
An agreed set of rules governing how data is transmitted between devices.
`Credited: Set of rules for communication`

FTP
File Transfer Protocol: rules for uploading and downloading files between computers over a network.

HTTP
HyperText Transfer Protocol: rules a browser uses to request web pages from a web server and transfer them.

HTTPS
HyperText Transfer Protocol Secure: HTTP with encryption, so data sent between browser and server cannot be read if intercepted.

Exam focus

This lesson is tested almost every year. Encryption appeared in 2021 Q10, 2022 Q12, 2023 Q8 and 2024 Q6; firewalls in 2022 Q12; strong passwords in 2021 Q9; and protocols, FTP and HTTPS in 2021 Q9, 2022 Q7 and 2023 Q8. 2024 Q11 asked for protection methods in context, and terms such as protocol and firewall turn up in matching questions. For encryption, two marks usually need what it does (scrambles data so it cannot be understood) and who can read it (only someone with the key). Never say encryption stops data being stolen. For a protocol, say it is a set of rules and give a detail, such as allowing different devices to communicate. When asked to 'expand' an acronym, spell every word correctly. For HTTP versus HTTPS, the key point is that HTTPS encrypts the data, so you must compare both sides. Always link protection methods to the scenario given.

`Unit 1 2024 Q11`	`[6]`	Legislation matching, online harms, protection methods
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q6`	`[13]`	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes
`Unit 1 2023 Q8`	`[9]`	Protocols, HTTPS, phishing, malware, encryption
`Unit 1 2022 Q12`	`[7]`	WAN, encryption, firewall, processor performance

? Check yourself

1. Explain how encryption protects data. [2]
2. Describe the purpose of a firewall. [2]
3. Explain how levels of access protect patient data in a hospital. [2]

Answers: 1. Data is scrambled/encoded using a key/algorithm (1) / so if it is intercepted or stolen it cannot be understood; only someone with the key can decrypt it (1). 2. Monitors/filters data entering and leaving a network (1) / blocks traffic that does not meet the security rules, e.g. stops hackers gaining access from the internet (1). 3. Staff are given access only to the data they need for their job, e.g. receptionists see appointments but not medical notes (1) / so confidential patient data is not seen or changed by unauthorised staff; read-only access prevents records being altered (1).

Cloud technology

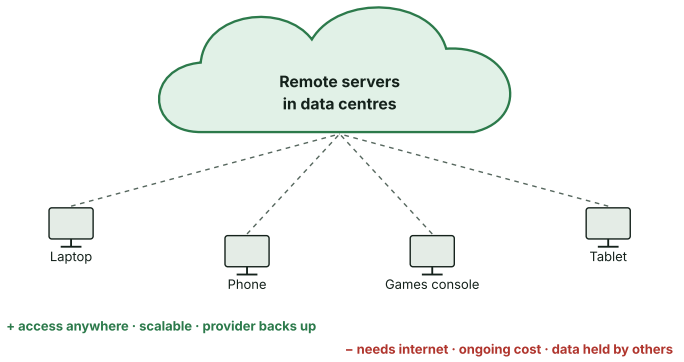
L20 FACT FILE

Specification: students should be able to

- Define the term cloud computing
- Describe the advantages of cloud computing for an organisation
- Describe the disadvantages of cloud computing for an organisation
- Describe the impact of cloud computing on gaming
- Describe the impact of cloud computing on file storage and file sharing, including collaborative tools

CCEA fact file: Fact File 17: Cloud Technology

Key words: cloud computing, remote server, data centre, on demand, subscription, pay as you go, scalability, downtime, bandwidth, carbon footprint, cloud gaming, game streaming, cloud storage, synchronisation, collaboration, version history



Cloud computing: devices reach remote servers over the internet

Cloud computing explained

Cloud computing means using computing resources, such as storage space and software, that are delivered on demand over the internet from remote servers. Instead of keeping files and programs on their own hard drives, organisations and individuals keep them on servers in large data centres owned by cloud providers. They can then reach them from any device with an internet connection. Familiar examples include OneDrive, Google Drive, iCloud and Microsoft 365. Organisations usually pay a monthly subscription or use a pay-as-you-go plan. The cloud is growing quickly because it is flexible: a business can start small and add more storage or users when it needs them. It has also changed everyday life for young people, from saving coursework online to streaming games and working on shared documents with classmates.

Advantages for organisations

For an organisation, the cloud can save money because it does not need to buy, house and maintain its own servers or many separate software licences. Pay-as-you-go plans mean it pays only for what it uses. The service is scalable, so storage and bandwidth can be increased or reduced quickly. Storage capacity is far larger than local hard drives. The provider applies software updates automatically and backs up data at regular intervals, so the organisation does not need its own backup devices or recovery procedures. Staff can access their files from anywhere and on any device, which supports home and hybrid working. If the office suffers a fire, flood or theft, the data is still safe. Using less equipment on site also reduces electricity use, which lowers the organisation's carbon footprint.

Disadvantages for organisations

The biggest weakness of the cloud is that it depends on the internet. If the broadband connection or the router fails, staff cannot reach their files or software. This is called downtime. Downtime can also happen if the provider has a fault or is carrying out maintenance, and the organisation cannot fix the problem itself. Some plans have a bandwidth allowance and charge extra if it is exceeded, and uploading large files can be slow. Security is another concern. The data is stored by a third party, and no provider can promise that it will never be hacked. The organisation is still responsible for protecting personal data under the UK GDPR and the Data Protection Act 2018. Subscription costs continue for as long as the service is used, and getting help from customer support can be slow or cost extra.

Impact on gaming

The cloud has changed gaming for both producers and players. Producers can sell games as downloads, so they save the cost of making discs and paying shops, and can offer lower prices. Players can download a game the moment it is released. Games kept on the producer's servers are much harder to copy illegally than discs, which helps protect sales. The internet also gives producers a worldwide market, and they can release patches, updates and demos straight away. With cloud gaming, also called game streaming, the game runs on powerful servers and video is streamed to the player's device. This means high-quality games can be played on cheap devices without upgrading hardware. However, players need fast broadband with low latency, streaming uses a lot of data, and access is lost if the subscription ends.

Impact on file storage and sharing

Cloud storage keeps files on high-capacity servers. Many apps save work automatically every few seconds, so a crash rarely means lost work. Files remain online until the user deletes them, and the provider keeps copies so it can recover files if a server fails, without the user needing any technical knowledge. Synchronisation means the latest version of a file appears on every device the user owns. Sharing is easier too: instead of emailing large attachments, a user shares a link and sets permissions to decide who can view or edit. Collaborative tools such as Google Docs and Microsoft 365 let several people edit the same document at the same time, wherever they are. Version history shows who made each change and allows an earlier version to be restored, and users can be sent alerts when a document changes.

T Key terms from the mark schemes

Cloud computing
Storing data and running software on remote servers that are accessed on demand over the internet, instead of on a local computer.
`Credited: Using remote servers over the internet`

Data centre
A building containing large numbers of servers that store data and run cloud services.

Pay as you go
A payment model where an organisation pays only for the cloud resources it actually uses.

Cloud gaming
Games run on remote servers and the video is streamed to the player's device over the internet.
`Credited: Game streaming`

Cloud storage
Keeping files on a provider's remote servers so they can be accessed from any device with an internet connection.
`Credited: Online storage`

Scalability
The ability to increase or decrease storage, bandwidth or users quickly as an organisation's needs change.
`Credited: Scalable`

Downtime
A period when a service is unavailable, for example because of an internet failure or provider maintenance.
`Credited: Service outage`

Synchronisation
Keeping copies of a file up to date so that a change made in one place appears on every device and for every user.
`Credited: Sync`

Collaborative tool
Cloud software that lets several people view and edit the same file at the same time, e.g. Google Docs or Microsoft 365.

Carbon footprint
A measure of the impact of an organisation's or person's activities on the environment.

Version history
A record of the changes made to a file that shows who made them and allows earlier versions to be restored.

Exam focus

Cloud questions are usually worth a few marks each but come up often. You may be asked to define cloud computing (2023 Q9, 2024 Q2). A two-mark definition needs two ideas: storing data or running software on remote servers, and accessing them over the internet. Benefits and risks for a business were asked in 2023 Q9 and 2024 Q2. Make each point specific and explain why it matters, for example 'needs an internet connection, so if the broadband fails staff cannot access files'. Cloud gaming was a quality of written communication question in 2019 Q7 and returned in 2023 Q9(c), so practise writing a balanced answer in full sentences that covers both benefits and drawbacks. Cloud storage terms appeared in 2022 Q2. Avoid vague answers such as 'it is cheaper' or 'it is easier' without saying why.

Unit 1 2024 Q2	[4]	Cloud definition, benefits and risks
Unit 1 2023 Q9	[6]	Cloud computing, business benefits, cloud gaming
Unit 1 2022 Q2	[3]	Optical, capacity, cloud storage terms
Unit 1 2019 Q7	[4]	Cloud computing benefits for gaming (QWC)

? Check yourself

1. Define the term cloud computing. [2]
2. A small accountancy firm in Newry keeps all its files on one ageing server in the office. State three benefits the firm would gain by switching to cloud computing. [3]
3. A solicitor's office in Armagh is moving its files to the cloud. Describe two disadvantages of cloud computing for the office. [4]

Answers: 1. Storing data and/or running software/applications (1) on remote servers accessed over the internet (1) / resources delivered on demand (1) – max 2 2. Any three: cost effective / pay as you go / no servers to buy (1) / scalable – storage can be increased or decreased (1) / very large storage capacity (1) / automatic software updates (1) / automatic backup and recovery by the provider (1) / access from anywhere on any device (1) / staff can collaborate on the same files (1) / reduced carbon footprint (1) 3. Two disadvantages, each identified (1) and described in context (1): relies on an internet connection (1) so if the broadband fails staff cannot open client files (1) / client data held by a third party (1) could be hacked, and the office must still meet UK GDPR (1) / provider outages or maintenance (1) mean files are unavailable and the office cannot fix it (1) / ongoing subscription costs (1) that continue every month (1) / bandwidth limits (1) with extra charges if exceeded (1) / slow customer support (1)

Legislation 1: Consumer Contracts, Copyright and Computer Misuse

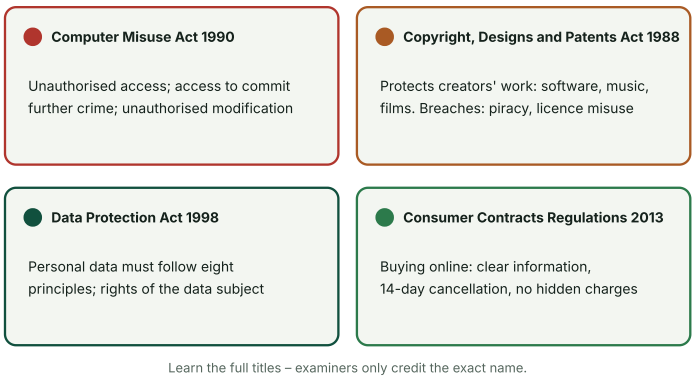
L21 FACT FILE

Specification: students should be able to

- Demonstrate knowledge and understanding of the Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013
- Demonstrate knowledge and understanding of the Copyright, Designs and Patents Act 1988
- Identify typical breaches of the Copyright, Designs and Patents Act 1988, including software piracy and software licensing infringements
- Demonstrate knowledge and understanding of the Computer Misuse Act 1990 and its offences
- Describe the terms hacker, virus and spyware and explain how they relate to the Computer Misuse Act 1990
- Apply knowledge of legislation to identify which law has been broken in a given scenario

CCEA fact file: Moral and Ethical

Key words: Consumer Contracts Regulations 2013, distance contract, cooling-off period, pre-ticked box, Copyright, Designs and Patents Act 1988, intellectual property, copyright, software licence, software piracy, plagiarism, Computer Misuse Act 1990, unauthorised access, hacker, malware, virus, spyware



Learn the full titles – examiners only credit the exact name.

Four laws examined in Unit 1 and what each covers

Laws that protect digital users

Digital technology lets us shop, create and communicate easily, but it also makes it easy to cheat customers, copy other people's work and break into computers. Three laws in this lesson deal with these problems. The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013 protect people who buy online, by phone or by mail order. They must be given clear information before paying, they get 14 days to change their mind about most goods, and they cannot be charged hidden extras. The Copyright, Designs and Patents Act 1988 protects intellectual property, such as software, music and images, so that only the owner can decide how it is copied or used. The Computer Misuse Act 1990 makes it a crime to access or change a computer system without permission. Together with data protection law, covered in the next lesson, these are the laws you need to name and apply.

Shopping at a distance

When you buy online you cannot see or handle the product, so the Consumer Contracts Regulations 2013 give you extra rights. Before you pay, the trader must give their identity and contact details, a description of the product, the full price including delivery, and how to cancel. Digital content must come with information about which devices it works with. For goods, the cooling-off period lasts 14 days from the day after delivery, and you do not need a reason to cancel. The refund must arrive within 14 days of the goods coming back or proof of posting. There are exceptions. You lose the right to cancel a download once it starts, as long as you agreed to that. Unsealed DVDs, games and software, and personalised items cannot be returned unless faulty. Pre-ticked boxes for extra payments are banned. Since 2025 the Digital Markets, Competition and Consumers Act 2024 has added further protection, such as banning fake reviews.

Copyright and software licences

Copyright protects original work automatically from the moment it is created. Under the Copyright, Designs and Patents Act 1988 a computer program is treated as a literary work, so its code cannot be copied without the owner's permission. When you 'buy' software you are really buying a licence: a legal agreement saying how many people or computers may use it and for what purpose. A school might buy a network licence for 30 machines or a site licence for the whole campus. Software piracy means illegally copying, downloading, sharing or selling software. Licence infringement includes installing software on more computers than allowed, sharing a licence key, using a student licence for business or carrying on after a subscription ends. Free software is not automatically free of copyright: freeware is still owned, while open-source licences allow the code to be changed and shared on stated terms. Breaches can lead to unlimited fines and, in serious cases, prison.

Computer misuse

The Computer Misuse Act 1990 was passed because, before it, hacking was hard to prosecute. It has since been updated. Section 1 makes unauthorised access to computer material an offence, even if nothing is changed. Section 2 covers unauthorised access with intent to commit a further crime, such as fraud or blackmail. Section 3 covers unauthorised acts that impair a computer, for example modifying or deleting data, spreading a virus, locking users out by changing their passwords, or launching a denial-of-service attack. Later additions cover acts that risk serious damage, for example to hospitals or power supplies (section 3ZA), and making or supplying hacking tools (section 3A). A hacker is someone who breaks into a system or its data without permission. A virus is malware that copies itself and damages files. Spyware secretly records a user's activity and sends it to someone else. Penalties rise with seriousness, from fines up to long prison sentences.

T Key terms from the mark schemes

Consumer Contracts Regulations 2013
The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013: protect people who buy online, by phone or by mail order by requiring clear information, giving a 14-day right to cancel and banning hidden extra charges.
`Credited: Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013`

Cooling-off period
The 14 days in which a consumer can cancel a distance purchase without giving a reason and get a refund; for goods it starts the day after delivery.
`Credited: cancellation period`

Copyright, Designs and Patents Act 1988
Protects the intellectual property of creators, including software, music, images and text, so it cannot be copied, shared, sold or adapted without the owner's permission.
`Credited: CDPA 1988`

Intellectual property
Original creative work, such as software, music, designs or writing, which belongs to the person or organisation that created it.

Software licence
A legal agreement that states how a piece of software may be used, including how many users or computers may run it.

Software piracy
Illegally copying, downloading, distributing or selling software without the copyright owner's permission.
`Credited: illegal copying of software`

Computer Misuse Act 1990
Makes it a criminal offence to access computer material without permission, to do so in order to commit another crime, or to carry out unauthorised acts that impair a computer, such as changing or deleting data.

Unauthorised access
Getting into a computer system, account or data without permission; an offence under section 1 of the Computer Misuse Act 1990.
`Credited: hacking`

Hacker
Someone who gets into a computer system, or the data held on it, without permission.

Virus
Malicious software that attaches itself to files or programs and copies itself, often damaging or deleting data.
`Credited: malware`

Spyware
Malicious software that secretly records a user's activity, such as keystrokes, and sends it to a third party.
`Credited: keylogger`

Exam focus

Questions on these laws have appeared in most recent papers. In 2025 Q9(a) candidates had to give the full title of the consumer law, and the examiners noted that many could not name it precisely, so learn 'Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013'. Scenario questions ask you to match a situation to the law broken (2024 Q11(a), 2025 Q9(b)): name the Act and its year, then link it to the detail in the scenario. Copyright has come up in 2022 Q9 and 2023 Q10, where marks were given for naming the Act and describing a breach such as installing software without a licence. 2019 Q1(d) and 2021 Q10 also tested consumer law and computer misuse. For a 'describe' question, make one point and then add detail or an example for the second mark. Common mistakes are mixing up the Computer Misuse Act with data protection and writing 'copywrite'.

`Unit 1 2025 Q9`	`[4]`	Consumer law title; data, computer misuse and copyright scenarios
`Unit 1 2024 Q11`	`[6]`	Legislation matching, online harms, protection methods
`Unit 1 2023 Q10`	`[5]`	Copyright, DPA principles, DPO and data controller
`Unit 1 2022 Q9`	`[8]`	RSI, monitor ergonomics, copyright, ICO, DPA principles
`Unit 1 2021 Q10(b)–(e)`	`[10]`	Cybercrime, access rights, DPA principles, encryption, legislation

? Check yourself

1. Give the full title of the legislation that gives people buying goods online a 14-day period in which to cancel their order. [1]
2. Describe two rights given to consumers by the Consumer Contracts Regulations 2013. [4]
3. Ciara buys a game download from an online store and starts playing it. Two days later she asks for a refund. Explain why the store does not have to give her one. [2]

Answers: 1. The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013 (1). Accept 'Consumer Contracts Regulations 2013'; the year is needed. 2. Any two rights, each with description (2 + 2): right to clear information before buying (1) / e.g. trader identity, total price including delivery, how to cancel (1); right to cancel within 14 days (1) / without giving a reason, starting the day after delivery (1); right to a refund (1) / within 14 days of returning the goods, including standard delivery (1); protection from hidden or extra charges (1) / pre-ticked boxes banned, must opt in (1); helplines charged at no more than basic rate (1) / so customers are not overcharged for support (1). 3. Ciara agreed to lose her right to cancel when the download started (1) / digital content cannot be returned once downloading or streaming has begun, so the right to cancel no longer applies (1).

Legislation 2: Data protection (UK GDPR and the Data Protection Act 2018)

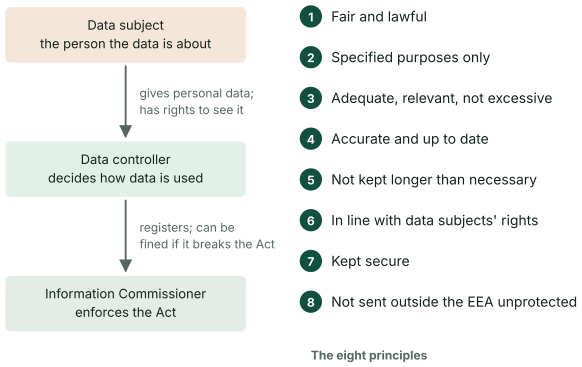
L22 FACT FILE

Specification: students should be able to

- Demonstrate knowledge and understanding of data protection law: the UK GDPR and the Data Protection Act 2018 (which replaced the Data Protection Act 1998 named in the specification)
- Demonstrate and apply knowledge and understanding of the data protection principles, including how they relate to the eight principles of the 1998 Act
- Describe the rights of the data subject
- Describe the responsibilities of the data controller, the Data Protection Officer and the Information Commission (formerly the ICO) in making sure the law is followed
- Describe measures organisations use to comply with data protection law

CCEA fact file: Moral and Ethical

Key words: UK GDPR, Data Protection Act 2018, personal data, special category data, data subject, data controller, data processor, Data Protection Officer, Information Commission, ICO, data protection principles, consent, subject access request, right to erasure, data breach



Data Protection Act roles and its eight principles

Protecting personal data

Personal data is any information about a living person who can be identified from it, such as a name, address, photo, email address or exam results. Organisations hold large amounts of it, and if it is lost or misused people can suffer fraud, identity theft or distress. Data protection law controls how organisations collect, store, use and share this data, and it gives people rights over their own information. The CCEA specification and many textbooks name the Data Protection Act 1998, but that Act was replaced in 2018. The law that applies in Northern Ireland today is the UK GDPR together with the Data Protection Act 2018, updated by the Data (Use and Access) Act 2025. It is enforced by the Information Commission (formerly the ICO). The main ideas have not changed: data must be collected fairly, used only for its stated purpose, kept accurate and secure, and deleted when it is no longer needed. Organisations that break the law can be fined up to £17.5 million or 4% of their worldwide turnover.

The principles in practice

The UK GDPR has seven principles. Personal data must be processed lawfully, fairly and in a transparent way, which means people are told what is happening to it. It must be collected for a clear purpose and not reused for something different. Only data that is adequate, relevant and necessary should be collected, so a club should not ask for details it does not need. Data must be accurate and kept up to date, and it must not be kept for longer than necessary. It must be kept secure using measures such as passwords, encryption, access rights and backups. Finally, the controller must be accountable, meaning it can show that it follows the rules. The old 1998 Act had eight principles that say almost the same things, plus a rule against sending data abroad without protection. In a scenario question, name the principle and then point to the exact detail in the scenario that breaks it.

Rights and responsibilities

The data subject is the person the data is about. They have the right to be told how their data is used, and the right to see it by making a subject access request, which is normally answered free of charge within one month. They can have wrong data corrected, ask for data to be deleted in some cases, object to direct marketing and claim compensation if they are harmed. The data controller is the organisation that decides how and why data is processed. It must follow the principles, pay a fee to the regulator, keep data secure, respond to requests and report serious breaches within 72 hours. Public bodies, including schools, must have a Data Protection Officer who advises staff and checks the rules are followed. The regulator enforces the law. On 30 September 2026, under the Data (Use and Access) Act 2025, the Information Commission (formerly the ICO) was replaced by the Information Commission, which does the same job: giving advice, investigating complaints and issuing fines.

Keeping data secure and accurate

Following the law is not just about paperwork. Organisations need practical measures. To keep data secure they use usernames and strong passwords, two-factor authentication, firewalls, anti-malware software and encryption, especially on laptops and memory sticks that might be lost. Access rights mean each member of staff can only see the data they need: a school receptionist might see a parent's phone number but not a pupil's medical notes. Regular backups protect against loss. To keep data accurate, organisations ask people to check their details, for example by sending a data collection sheet home at the start of each school year. Staff training is important, because many breaches are caused by simple mistakes such as sending an email to the wrong person or leaving a screen unlocked. Organisations should also have clear policies on how long data is kept and how it is securely deleted or shredded.

T Key terms from the mark schemes

UK GDPR
The UK version of the General Data Protection Regulation, which, with the Data Protection Act 2018, sets the rules for processing personal data in the UK.
`Credited: General Data Protection Regulation`

Data Protection Act 2018
UK law that works alongside the UK GDPR to control how personal data is collected, stored and used; it replaced the Data Protection Act 1998.
`Credited: Data Protection Act`

Personal data
Any information about a living person who can be identified from it, such as their name, address, photo or email address.

Special category data
Personal data that needs extra protection because its misuse could cause discrimination or harm, e.g. health, religion or ethnic origin.
`Credited: sensitive personal data`

Data subject
The living individual whom the personal data is about.

Data controller
The person or organisation that decides why and how personal data is processed and is responsible for following data protection law.

Data Protection Officer
A person who advises an organisation on data protection law, monitors that it is followed and acts as the contact for the regulator; required for public bodies such as schools.
`Credited: DPO`

Information Commission
The independent UK regulator that enforces data protection law, investigates complaints and can issue fines; it replaced the Information Commissioner on 30 September 2026.
`Credited: Information Commissioner / ICO`

Subject access request
A request by a data subject to see a copy of the personal data an organisation holds about them; normally answered free within one month.
`Credited: SAR`

Data breach
A security incident in which personal data is lost, destroyed, changed or disclosed without authorisation.

Exam focus

Data protection appears in almost every Unit 1 paper. You may be asked to state principles (2021 Q10(d), 2022 Q9(c), 2023 Q10(b)), describe the data controller and Data Protection Officer (2023 Q10(c)), describe the regulator's role (2022 Q9(c)), or recall data subjects' rights (2019 Q1(e)). Scenario questions such as 2024 Q11 and 2025 Q9(b) ask you to pick the right law. Older papers and mark schemes use the 1998 Act's wording and the title 'Information Commissioner' or ICO, which examiners still accept; either the 1998 principles or the current UK GDPR principles are acceptable if stated clearly. When asked to 'state' a principle, give it in full, for example 'kept no longer than necessary', not just 'time'. When applying a principle, link it to the scenario. A common mistake is confusing the data subject with the data controller, or describing the Computer Misuse Act when the question is about personal data.

Unit 1 2025 Q9	[4]	Consumer law title; data, computer misuse and copyright scenarios
Unit 1 2024 Q11	[6]	Legislation matching, online harms, protection methods
Unit 1 2023 Q10	[5]	Copyright, DPA principles, DPO and data controller
Unit 1 2022 Q9	[8]	RSI, monitor ergonomics, copyright, ICO, DPA principles
Unit 1 2021 Q10(b)–(e)	[10]	Cybercrime, access rights, DPA principles, encryption, legislation

? Check yourself

1. What is meant by the term 'data subject'? [1]
2. State four principles of data protection law. [4]
3. A shop in Ballymena stores customers' card details on an unencrypted laptop, which is stolen. State which data protection principle has been broken and describe one measure that would have prevented the breach. [3]

Answers: 1. The living individual whom the personal data is about (1). 2. Any four (1 each), current or 1998 wording: processed lawfully, fairly and transparently / collected for a specified purpose and not used for other purposes / adequate, relevant and not excessive (limited to what is necessary) / accurate and kept up to date / not kept longer than necessary / kept secure (integrity and confidentiality) / processed in line with the rights of the data subject (1998) / not transferred outside the UK/EEA without adequate protection (1998) / accountability. 3. Data must be kept secure / integrity and confidentiality principle (1); any one measure described (2): encrypt the laptop's storage (1) so data cannot be read if it is stolen (1) / do not store card details on portable devices (1), keep them on a secure server or use a payment provider (1) / password-protect the laptop with strong passwords (1) so a thief cannot log in (1).

Ethics, employment and health and safety

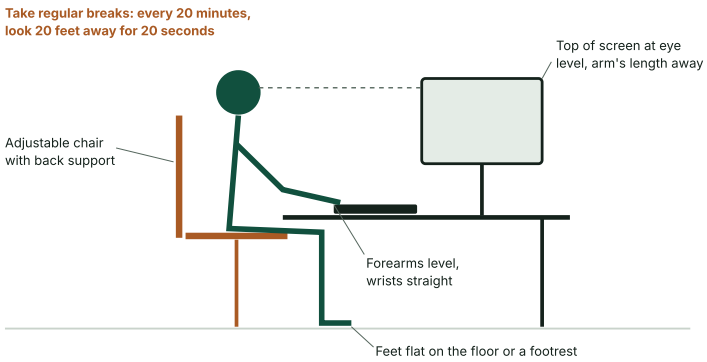
L23
FACT FILE

Specification: students should be able to

- Describe the ethical impact of technology on society, referring to internet misuse, access to personal information, social media misuse, the implications of GPS and tracking, and concerns about the security of personal data
- Describe the impact of digital technology on employment, including increased job opportunities in the digital technology and computing sector, job displacement, changes in work patterns and the need for upskilling
- Demonstrate understanding of digital technology related health and safety issues, including RSI, back strain and eye strain
- Identify the measures that both the employee and employer should take to promote good health and safety practice in the workplace, including the Health and Safety (Display Screen Equipment) Regulations 1992 (in Northern Ireland, the Health and Safety (Display Screen Equipment) Regulations (Northern Ireland) 1992)

CCEA fact file: Moral and Ethical, Fact File 20: Changes in employment, Health and Safety

Key words: ethics, internet misuse, acceptable use policy, phishing, social media misuse, cyberbullying, trolling, GPS, tracking, privacy, job displacement, upskilling, teleworking, RSI, back strain, eye strain, ergonomic, Display Screen Equipment Regulations (NI) 1992



A correctly set-up computer workstation

Technology and society

Ethics is about what is right and wrong, not just what is legal. Digital technology brings huge benefits, but it also raises ethical questions. The internet can be misused for crimes such as phishing, stealing account details and piracy, and for spreading fake news or hateful content. Huge amounts of personal information are collected through loyalty cards, card payments, cookies, phone location and CCTV, often without people really noticing. Social media can be misused for cyberbullying, trolling, fake profiles and sharing images without consent. GPS and tracking help with navigation and safety, but can invade privacy. Technology also changes jobs: it creates new ones in the digital sector, replaces others, changes when and where people work, and means everyone needs to keep learning. Finally, spending long hours at a computer can cause health problems such as RSI, back strain and eye strain, so employers and employees both have responsibilities to work safely.

Misuse, privacy and tracking

Internet misuse includes crimes and behaviour that harms others or wastes time. Phishing uses fake but official-looking messages to trick people into giving details, and stolen passwords are then used to buy goods or empty accounts. Plagiarism and illegal downloading take credit or profit from other people's work. Employers deal with staff misuse through an acceptable use policy. On social media the main misuses are cyberbullying, trolling, hate speech, fake profiles used for identity theft and oversharing personal details. Mark schemes treat trolling and cyberbullying as different points: bullying targets one person repeatedly, while trolling aims to provoke anyone. GPS uses satellites to work out a device's exact position. It helps parents, delivery firms, emergency services and carers of people with dementia, but tracking without consent invades privacy and could allow stalking. When data is stored in the cloud on servers abroad, it may be protected by weaker laws, which raises concerns about its security.

Jobs and work patterns

Digital technology has created many new jobs, such as software engineers, web and app designers, game developers, cyber security and digital forensics analysts, and database administrators. Northern Ireland has a growing technology sector, especially around Belfast. At the same time, job displacement happens when technology takes over work people used to do: robots assemble and paint cars, self-service tills replace checkout staff, and online banking has led to branch closures. Work patterns have changed too. Teleworking means using ICT, such as laptops, broadband, cloud apps and video calls, to work from home. It saves travel time and reduces pollution, but it can blur the line between home and work. Global companies and 24-hour call centres mean more shift work and staff spread across time zones. Because technology changes quickly, workers need upskilling, which is training in new skills. Good employers provide training because it improves productivity, staff morale and staff retention.

Staying healthy at a computer

Repetitive strain injury (RSI) is caused by repeating the same small movements, such as typing or clicking, for long periods. It causes pain and stiffness in the hands, wrists, arms or shoulders. Back strain comes from poor posture and unsuitable chairs. Eye strain is caused by staring at a screen for too long, especially with glare or poor lighting, and leads to headaches and blurred vision. Employees should take regular breaks, sit correctly, adjust their chair and screen and use wrist rests. Employers must provide a safe working environment under the Health and Safety at Work (Northern Ireland) Order 1978. Under the Health and Safety (Display Screen Equipment) Regulations (Northern Ireland) 1992, they must assess workstations, plan breaks or changes of activity, train staff and provide eye tests for regular screen users. They should supply adjustable chairs, footrests, ergonomic keyboards and mice, anti-glare screens and good lighting. Other safety measures include tidy cables, PAT testing of equipment, keeping food and drink away from computers and having CO_2 fire extinguishers.

Key terms from the mark schemes

Phishing
Using fake emails, texts or adverts that look official to trick people into giving away personal details such as passwords.

Acceptable use policy
Rules that employees or pupils agree to that set out how they may use an organisation's computers and internet.
`Credited: AUP`

Trolling
Deliberately posting provocative or offensive messages online to upset people or start arguments; credited separately from cyberbullying.

Cyberbullying
Using technology, such as social media or messages, to repeatedly hurt, threaten or embarrass a particular person.

Fake news
Deliberately false or misleading information that is spread online, often made to look like real news.

GPS
Global Positioning System: uses signals from satellites to work out the exact location of a device in real time.
`Credited: Global Positioning System`

Job displacement
When jobs are lost or changed because digital technology now does work that people used to do.

Automation
Machines or software carrying out tasks that were previously done by people.

Upskilling
Training workers in new skills, especially digital skills, so they can keep up with changes in technology.
`Credited: retraining`

Teleworking
Using ICT, such as broadband, laptops and video conferencing, to work from home instead of at the workplace.
`Credited: remote working / working from home`

Repetitive strain injury (RSI)
Pain or injury in the hands, wrists, arms or shoulders caused by repeating the same movements, such as typing, for long periods.
`Credited: RSI`

Ergonomic
Designed to fit the user and reduce physical strain, e.g. adjustable chairs, ergonomic keyboards and wrist rests.

Exam focus

This lesson often provides the extended-writing questions, which are marked for quality of written communication. 2021 Q8(a) asked about social media misuse and 2023 Q11(a) about the impact on employment, both for 6 marks. For these, cover several different points, explain each with an example, and write in organised paragraphs. Health and safety appeared in 2022 Q9(a) (RSI and monitor set-up), 2024 Q9(b) (health risks) and 2025 Q10, where examiners found answers about posture and ergonomics were often poorly targeted. Read whether the question asks about the employee or the employer and answer for that person: 'take a break' is an employee measure; 'provide adjustable chairs' is an employer measure. Match the measure to the problem asked about, so a wrist rest helps RSI, not eye strain. 2024 Q11(b) covered online harms, and 2019 Q1(f)–(g) and the key-term matching questions (2021 Q11, 2024 Q10) tested eye strain, GPS and job displacement.

`Unit 1 2025 Q10`	`[4]`	RSI/posture causes and ergonomic prevention
`Unit 1 2024 Q11`	`[6]`	Legislation matching, online harms, protection methods
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q9`	`[7]`	Input devices, health risks, gaming and internet risks
`Unit 1 2023 Q11(a)`	`[6]`	Impacts on employment (QWC)

? Check yourself

1. Describe two ethical concerns about the use of GPS tracking. [4]
2. Ballymena Academy's ICT co-ordinator is writing a talk for Year 10 about how the internet is misused. Describe three different examples of misuse that the talk could include. [6]
3. Describe two ways in which a supermarket can collect personal data about its customers without them fully realising it. [4]

Answers: 1. Two concerns, each identified (1) and described (1): a person's location can be tracked without their consent (1) which invades their privacy (1); could be used for stalking or controlling someone (1) by following their live location (1); employers may monitor staff constantly (1), even outside working hours (1); apps may sell location data to advertisers (1) without users realising (1); geotagged posts can reveal home or school (1), putting people at risk (1). 2. Three ways, each identified (1) and described (1): phishing (1) – fake official-looking emails/adverts trick people into giving personal details (1); theft of usernames and passwords (1) – used to access accounts and make purchases (1); piracy (1) – illegally downloading copyrighted music, films or software (1); plagiarism (1) – copying work and claiming it as your own (1); spreading fake news or hateful/violent content (1) – misleads or harms people (1); using the dark web to access illegal material (1); employees using work internet for personal use (1) – wastes time, leading to acceptable use policies (1). 3. Two ways, each identified (1) and described (1): loyalty cards (1) record what each customer buys and when, used for targeted advertising (1); card payment records (1) show shopping patterns, which may be shared or sold (1); online shopping accounts/cookies (1) track browsing and purchases (1); CCTV (1) records customers in store (1); app/Wi-Fi location data (1) shows when customers visit (1).

Digital applications and extended-answer technique

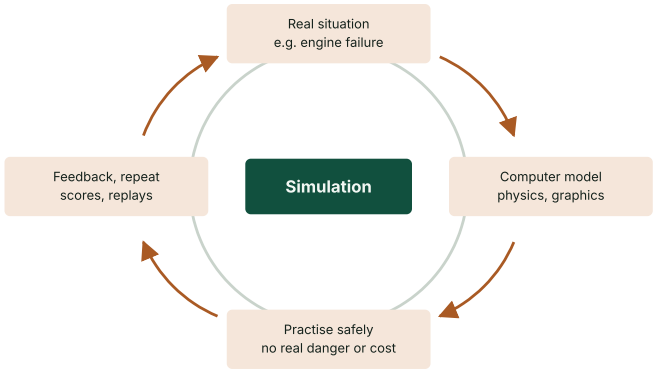
L24 FACT FILE

Specification: students should be able to

- Describe the main features of gaming applications, simulations and mobile phone applications
- Describe how gaming applications, simulations and mobile phone applications can be used to support education and training, social interactions and work practices
- Describe how virtual reality is used to create realistic simulations for training
- Evaluate the impact of online banking on customers and on banks
- Evaluate the impact of online training on employees and employers
- Evaluate the impact of e-commerce on customers and on businesses

CCEA fact file: Digital Applications

Key words: gaming application, cloud gaming, leaderboard, gamification, immediate feedback, simulation, computer model, virtual reality, mobile app, app store, online banking, two-factor authentication, phishing, online training, automatic tracking, e-commerce, evaluate, quality of written communication



How a simulation supports training: model, practise safely, get feedback, repeat

Digital applications in everyday life

Digital applications are programs that help people learn, train, stay in touch, work, bank and shop. Gaming applications use rules, levels, points, leaderboards and immediate feedback to motivate players, and online multiplayer lets people play and talk together in real time. Simulations use a computer model, built from data and mathematical rules, to copy how a real object or situation behaves, so people can test designs or practise dangerous tasks safely. Virtual reality takes this further by placing the user inside a 3D world through a headset. Mobile phone apps are always with the user and use features such as notifications, GPS and the camera to support learning, communication and work on the move. Online banking, online training and e-commerce have changed how we manage money, gain skills and buy goods. Each brings clear benefits, such as 24/7 access and convenience, but also drawbacks, such as fraud risk, the need for internet access and digital skills, and less face-to-face contact. In the exam you must be able to describe the features and uses of the first three and evaluate the impact of the last three on both users and organisations.

Games, simulations and apps at work and school

Games suit training because they are built in levels: each level brings a new challenge and you cannot skip ahead, so skills are built step by step. Immediate feedback shows the learner what went well and what to try next, and role play lets them make decisions and reflect on the results without real consequences. Leaderboards and badges, often called gamification when they are added to non-game tasks, encourage people to keep improving. Simulations are used where real practice would be dangerous, expensive or rare. A trainee pilot can face engine failure or a thunderstorm in a flight simulator, with every action recorded for feedback, and no fuel is used and no aircraft is taken out of service. Surgeons can rehearse an operation on a 3D virtual patient made from scan data. Mobile apps let teachers share video, audio and e-books, let pupils reach their work through the cloud at home and school, let managers message groups of staff, and let schools tell parents about events. Good apps load quickly, keep personal data secure, can be personalised and are kept up to date by their developers.

Evaluating online banking and online training

Online banking lets customers check balances, view statements, pay bills, transfer money and set up standing orders and direct debits at any time, even from abroad, and Faster Payments moves money almost instantly. However, customers need a device, an internet connection and the skills to use it, so some older or less confident people are excluded. Fraud is a real worry: phishing emails and fake texts try to steal login details. Some customers miss face-to-face help. For banks, online services cut costs because fewer branches and staff are needed, and e-statements save paper and postage, but banks must protect huge amounts of personal data with complex, constantly updated security and must employ specialist staff. Online training lets employees learn at their own pace and at a convenient time, saves travel, lets staff in different places train at once and records progress automatically. Multimedia and VR simulations make it engaging and realistic. On the other hand, support may be limited to FAQs or email, learners can feel isolated, it needs self-motivation, deadlines are hard to manage and a poor connection can stop learning altogether.

Evaluating e-commerce

E-commerce means buying and selling goods and services over the internet. For customers it is convenient: they can shop at any time without travelling, which especially helps people with disabilities, parents of young children and those living far from town centres. They can read reviews, compare prices, choose from a far wider range than a local shop stocks and pre-order popular items. For businesses, there is no need for expensive high-street premises or checkout staff, so savings can be passed on as lower prices, and a website can reach customers worldwide. The drawbacks are also real. Customers cannot see or try goods before buying, so items may not match the photos and must be posted back. They wait for delivery and often pay extra for it, and fake websites may be set up to steal card details. Secure payment services and checking for HTTPS reduce this risk. In the UK, the Consumer Contracts Regulations 2013 usually give online shoppers 14 days after receiving goods to cancel the order. High-street shops may close as more people shop online, affecting town centres and jobs.

T Key terms from the mark schemes

E-commerce
The buying and selling of goods and services over the internet.
`Credited: Online shopping; electronic commerce`

Online banking
Using a bank's secure website or app to manage an account, for example viewing statements, paying bills and transferring money.
`Credited: Internet banking`

Online training
Learning delivered over the internet, often with multimedia and automatic progress tracking, that learners can complete at their own pace.
`Credited: Web-based training; e-learning`

Gaming application
An electronic game played on a console, computer or mobile device, often online with other players.

Gamification
Adding game features such as points, badges and leaderboards to a non-game activity to increase motivation.

Simulation
A computer model, built from data and mathematical rules, that copies how a real object or situation behaves so it can be tested or practised safely.
`Credited: Computer model`

Virtual reality
A computer-generated 3D environment that the user enters using a headset and sometimes gloves or controllers, stimulating senses such as sight, hearing and touch.
`Credited: VR`

Mobile app
A software application designed to run on a smartphone or tablet, usually downloaded from an app store.
`Credited: Mobile phone application`

Phishing
Fake emails, texts or websites that pretend to be from a trusted organisation in order to trick people into giving personal or login details.

Two-factor authentication
Checking a user's identity using two different methods, such as a password plus a code sent to their phone.
`Credited: 2FA`

GUI
Graphical user interface: lets users interact using windows, icons, menus and a pointer, or touch on a mobile app.
`Credited: WIMP interface`

Quality of written communication
Credit in extended answers for organised writing, correct spelling and grammar, and accurate use of specialist terms.
`Credited: QWC`

Exam focus

This topic is almost always tested through 6-mark extended questions that also assess quality of written communication: advantages of online training (2019 Q9), disadvantages of online shopping for customers (2022 Q6) and disadvantages of online banking to the customer (2025 Q12). Shorter questions have asked for drawbacks of online learning (2023 Q11(b)), risks of gaming (2024 Q9) and matching key terms such as e-commerce (2021 Q11). Answers are marked in levels: roughly Level 1 (1-2 marks) for a few basic points, Level 2 (3-4) for some developed points and Level 3 (5-6) for several well-developed, accurate points in an organised answer using specialist terms. Read the question carefully: if it asks about the customer, points about the bank or business earn nothing, and if it asks for disadvantages, advantages are not credited. Develop every point with a reason or consequence. Words such as 'faster', 'easier' or 'cheaper' on their own are not credited, so explain what is faster and why it matters. Spell key terms correctly and write in paragraphs.

Unit 1 2025 Q12	[6]	Disadvantages of online banking to the customer (QWC)
Unit 1 2024 Q9	[7]	Input devices, health risks, gaming and internet risks
Unit 1 2023 Q11(b)	[2]	Drawbacks of online learning
Unit 1 2022 Q6	[6]	Disadvantages of online shopping for customers (QWC)
Unit 1 2021 Q11	[8]	Key term matching

? Check yourself

1. State what is meant by e-commerce. [1]
2. State two features of gaming applications that help to motivate learners. [2]
3. Describe how gaming applications support social interaction. [2]

Answers: 1. Buying and selling goods or services over the internet (1). 2. Any two from: leaderboards (1) / points or scores (1) / badges or rewards (1) / levels of increasing difficulty (1) / immediate feedback (1) / multimedia such as animation and sound (1). 3. Any one developed point, e.g. online multiplayer lets players compete or co-operate with others anywhere in the world (1) and talk in real time using headsets and microphones or text chat (1). Accept joining online gaming groups/communities and collaborating to solve problems.